
Matroidal degenerations of Prym varieties

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Abstract. Let C_0 be a stable curve of which its irreducible components have arithmetic genus 0. The universal deformation $\mathcal{C} \rightarrow \text{Def}(C_0)$ gives a matroidal family of Jacobians $\mathcal{JC}^* \rightarrow \text{Def}(C_0)^*$ where the matroid is given by the cographic matroid of the dual graph of the central fiber $\Gamma(C_0)$. In this master's thesis, we will establish a similar connection for degenerations of Prym varieties. If (C_0, ι_0) is a pair of a stable curve with an admissible involution, the universal deformation of the pair gives a family of Prym varieties $P(C, \iota)^* \rightarrow \text{Def}(C_0, \iota_0)^*$. If the involution on $\Gamma(C_0)$ fixes no vertices, one may associate a natural signed graph Σ_ι , and if this signed graph is tangled, the monodromy cone of the degeneration is matroidal with respect to the cosigned-graphic matroid $M^*(\Sigma_\iota)$. We also show how to assign to each tangled signed graph Σ an admissible double cover of stable curves such that the monodromy cone associated to the induced degeneration of Prym varieties is matroidal with matroid $M^*(\Sigma)$. Using this, we construct examples of matroidal degenerations of Prym varieties of which the underlying regular matroid is not cographic. By applying the main theorem of [EGFS25b], this will show that the integral Hodge conjecture fails for a very general point in the Prym locus $\mathcal{P}_g \subseteq \mathcal{A}_g$ for $g \geq 6$.

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1. INTRODUCTION

1.1. Matroids and degenerations of abelian varieties. Degenerations of principally polarized abelian varieties $X^* \rightarrow B^*$ come with rich combinatorial structure through the lens of monodromy, which measures how the homology classes of a fiber change when running around a singularity. The combinatorial structure is sometimes encoded by matroids, which are devices that track independence, similar to linear independence of a finite set of vectors in a vector space.

Recently, the connection between matroids and degenerations of abelian varieties has been explored in a new and interesting way by Engel, de Gaay Fortman and Schreieder [EGFS25b], to obtain obstructions the integral Hodge conjecture for one-cycles on abelian varieties. Their main theorem, [EGFS25b, Thm. 1.6], gives a criterion for the failure of the algebraicity of multiples of the minimal class on a very general fiber of the family. They apply their criterion to prove that the minimal class is not algebraic on a very general principally polarized abelian variety of dimension g when $g \geq 4$, and on the intermediate Jacobian of a very general cubic threefold, see [EGFS25b, Thm 1.1 & 1.3]. One may wonder whether the criterion in [EGFS25b, Thm 1.6] may be applied in other situations, and thus find other examples of abelian varieties that fail the integral Hodge conjecture. With this question in mind, the main purpose of this thesis is to compute the matroids arising from degenerations of Prym varieties.

Let us first consider the well-known case of degenerations of Jacobians. To any stable curve C_0 one may associate a degeneration of Jacobians $\mathcal{J}^* \rightarrow \text{Def}(C_0)^*$. The monodromy of such degeneration can be calculated using graph theory: it is given by the so-called cographic matroid of the dual graph $\Gamma(C_0)$ of the central fiber (see Prop. 4.17 for details). Conversely, one can associate a degeneration of Jacobians to any cographic matroid, simply because any graph defines a stable curve. In other words,

$$\{\text{degenerations of Jacobians}\} \longleftrightarrow \{\text{cographic matroids}\}$$

Another example is worked out in Gwena's thesis [Gwe04]. To the Segre cubic threefold Y_0 , one may associate a degeneration of intermediate Jacobians $IJY^* \rightarrow B^*$. The monodromy of this degeneration is given by the famous matroid R_{10} , which is not cographic. Using this fact, one can show that very general cubic threefolds are not rational. The calculation in Gwena's thesis involves an intermediate step where he deforms the family of intermediate Jacobians to a family of Prym varieties. We will investigate these thoroughly in this thesis.

1.2. Matroids arising from degenerations of Prym varieties. Prym varieties are principally polarized abelian varieties which are naturally assigned to étale double covers of smooth curves $P(f : C \rightarrow C')$. The central fiber of a degenerations of Prym varieties gives double covers of graphs, to which Zaslavsky associated matroids in [Zas82] by developing the theory of signed graphs. A signed graph is a pair $\Sigma = (G, \sigma)$ consisting of a graph G and a sign function $\sigma : E(G) \rightarrow \{\pm 1\}$ sending an edge of the graph to a sign. The matroids Zaslavsky introduced are named signed-graphic matroids

and their duals are called cosigned-graphic matroids. Given a double cover of graphs $G \rightarrow G'$, the idea of the sign function is that it remembers the double cover structure by just looking at the base G .

Now given an admissible double cover of stable curves $C_0 \rightarrow C'_0$, one may associate a degeneration of Prym varieties $P^* \rightarrow B^*$ (see Remark 5.4). The central fiber gives a double cover of graphs $\Gamma(C_0) \rightarrow \Gamma(C'_0)$ and we associate a signed graph Σ_ι to this double cover (see Section 7). It is known that the (co)signed-graphic matroid of any signed graph is near-regular, that is, they are realizable over all fields of characteristic not equal to 2 (see [Zas82, Thm. 8B.1]). We provide a canonical *near-integral realization* (see Definition 4.6) of $M^*(\Sigma_\iota)$, which in some cases (when the signed graph is tangled or balanced, see Definitions 6.3 and 6.13) is an integral realization:

Theorem 1.1. *Let G be graph with an admissible involution ι acting freely on the vertices. Then the linear map*

$$(1) \quad \begin{aligned} \varphi : E(G)/\iota &\rightarrow (H_1(G, \mathbb{Z})^-)^\vee \\ [e] &\mapsto x_e \end{aligned}$$

by restricting coordinate projections is a near-integral realization of $M^(\Sigma_\iota)$. Moreover, if Σ_ι is balanced or tangled, then $M^*(\Sigma_\iota)$ is regular, and (1) is an integral realization.*

The main result of the master's thesis is then the following theorem:

Theorem 1.2. *Let C_0 be a stable curve with an admissible involution ι_0 fixing no vertices of the dual graph. Assume that Σ_ι is balanced or tangled. Then the monodromy cone of the associated degeneration of Prym varieties $P^* \rightarrow (\Delta^*)^{k-j}$ is matroidal with underlying regular matroid $M^*(\Sigma_\iota)$. If all irreducible components of C_0 are of genus zero, then the degeneration is matroidal with regular matroid $M^*(\Sigma_\iota)$.*

Conversely, given a tangled signed graph Σ , we associate a degeneration of Prym varieties so that its monodromy cone is matroidal with matroid $M^*(\Sigma)$:

Theorem 1.3. *Given any tangled signed graph Σ , there is a degeneration of Prym varieties such that its monodromy cone is matroidal with regular matroid $M^*(\Sigma)$.*

Thus, in a sense:

$$\{\text{degenerations of Pryms}\} \longleftrightarrow \{\text{cosigned-graphic matroids}\}$$

Applying this to Gwena's example, the signed graph becomes $-K_5$ which is the complete graph on 5 vertices with all signs negative. By a theorem of Zaslavsky, the cosigned-graphic matroid of $-K_5$ is the matroid R_{10} .

Recall [EGFS25b, Thm. 1.6], which implies that if $X^* \rightarrow B^*$ is a matroidal family of principally polarized abelian varieties of dimension $g \geq 4$ with underlying matroid not cographic, then a very general fiber of this family fails the integral Hodge conjecture. Applying this theorem to suitable families of Prym varieties, we obtain the following:

Corollary 1.4. *For $g \geq 6$, a very general point on the Prym locus $\mathcal{P}_g \subseteq \mathcal{A}_g$ fails the integral Hodge conjecture.*

1.3. Historical remarks and comparisons. Even though the Torelli map

$$\begin{aligned}\mathcal{M}_g &\rightarrow \mathcal{A}_g, \\ C &\mapsto (J(C), \Theta)\end{aligned}$$

always extends to a morphism $\overline{\mathcal{M}}_g \rightarrow \overline{\mathcal{A}}_g^V$ to the second Voronoi compactification $\overline{\mathcal{A}}_g^V$, the Prym map from the moduli space of étale double covers of curves $\mathcal{R}_g$

$$\begin{aligned}\mathcal{R}_g &\rightarrow \mathcal{A}_{g-1}, \\ (f : C \rightarrow C') &\mapsto (P(f), \Xi)\end{aligned}$$

is known, since the work of Friedman-Smith [FS86], to sometimes fail extending to $\overline{\mathcal{R}}_g \rightarrow \overline{\mathcal{A}}_g^V$. Degenerations of Prym varieties have subsequently been studied throughout the years by various mathematicians, including [FS86], [ABH01], [Gwe04], [CMGHL15]. By work of Alexeev, Birkenhake and Hulek [ABH01] and Vologodsky [Vol01], the indeterminacy locus of the Prym map to the second Voronoi compactification was completely classified and consequently named the Friedmann-Smith locus. An attempt for the indeterminacy of the Prym map to the first Voronoi compactification was made in [CMGHL15]. A recurring theme throughout is that the monodromy bilinear forms of the degenerations were computed using the character lattice of the torus part of the semi-abelian variety at the central fiber. Our approach will, however, be of a more combinatorial flavour. From this angle, one sees that the monodromy is related to theory of signed graphs.

In a sense, the connection between signed graphs and tropical Prym varieties was established in [RZ23]. We independently found this connection in specific cases, and when finding the article, it allowed us to generalize our constructions.

1.4. Overview. We will start by reviewing all preliminaries, which will occupy Chapters 2, 3 and 4. All results here are known but sometimes hard to find in the literature. In Chapter 2 we will study involutions of curves and construct families of double covers of curves $(\mathcal{C}, \iota) \rightarrow B$ using deformation theory. We will then touch on the monodromy of curves of higher dimensional bases which will set the stage for the monodromy weight filtration in later chapters.

Chapter 3 is meant as a quick recollection of abelian varieties with a specific view towards Jacobians and Prym varieties. The monodromy weight filtration of principally polarized abelian varieties is introduced.

In Chapter 4 we will first recall the basics of matroids and then show how to relate degenerations of Jacobians to cographic matroids. Using the same strategy, we will extend our reasoning to Prym varieties in Chapter 5. We will use the families of double covers of curves $(\mathcal{C}, \iota) \rightarrow B$ constructed in Chapter 2 to construct degenerations of Prym varieties $P^* \rightarrow B^*$. The monodromy weight filtration of such a degeneration will be computed and we will then give a completely combinatorial manner to describe the monodromy bilinear forms in terms of the double cover of the dual graphs of the central fiber (see Theorem 5.7 and 5.17).

In Chapter 6, we will introduce the necessities needed for signed graphs and in Chapter 7, we will show the promised connection of Prym varieties to signed graphs. Theorem 5.17 provides us with the realization describing the monodromy of the degenerations constructed in the preceding chapters, so what is left to do is to investigate if this is a realization of certain matroids. We are forced to introduce a weaker notion of matroidal families as it can happen that our matroid is not regular (see Example 7.10). We then give a wealth of examples of matroidal families of Prym varieties, which we will then use for applications to the integral Hodge conjecture for abelian varieties. In the last chapter, we will compute the matroid R_{10} which arises from Gwena's example by hand, without using the theory of signed graphs.

The new material can be found in Chapters 5, 7 and 8.

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2. CURVES

Let us start with stating all the preliminaries of curves, and families of them, which are needed in later chapters.

Definition 2.1 ([ACG⁺11, X. Nodal curves]). *A curve will mean a reduced separated finite type $\mathbb{C}$ -scheme whose irreducible components all have dimension 1. The analytification of such schemes will also be called curves. By the genus of a curve, we will always mean the arithmetic genus. If C is a curve with $e \in C(\mathbb{C})$ a closed point, we*

say e is a node if we have an isomorphism

$$\hat{\mathcal{O}}_{C,e} \cong \frac{\mathbb{C}[[x,y]]}{(xy)}.$$

The normalization $\nu : C^\nu \rightarrow C$ gives exactly two preimages and the images of two neighbourhoods around these points in C are called the branches at e . A nodal curve is a curve with its singularities at worst nodal. We say a nodal curve is stable if it is connected and

$$2g_i - 2 + \#\{\text{special points of } C_i^\nu\} > 0$$

where C_i^ν is a connected component of the normalization of C and the special points are the preimages of nodes. Equivalently, a curve is stable if and only if its automorphism group is finite.

When we work with analytic spaces, a point e on a curve being a node is equivalent to saying e has an open neighbourhood U such that U can be identified with an open subset in $\{xy = 0\} \subseteq \mathbb{C}^2$.

Involutions of curves will play an important role in later chapters. The following is a lemma which we will use for analyzing different types of nodes under an involution. It is a well-known result of which we could not find a proof (see for example [ACG⁺11, pp. 191 eq. (4.13)]).

Lemma 2.2. *Let C be a stable curve and $f \in \text{Aut}(C)$ and nodes e, e' such that $f(e) = e'$. Then for suitable analytic coordinates around e and e' , the automorphism acts locally as $f : (x, y) \mapsto (ax', by')$ where a and b are n 'th roots of unity with n the order of f .*

Proof. Say $\frac{\mathbb{C}[[x,y]]}{(xy)}$ and $\frac{\mathbb{C}[[x',y']]}{(x'y')}$ are the completed local rings around e and e' respectively. Let C^ν be the normalization of C and denote by p_1, p_2 and p'_1, p'_2 the special points obtained from the respective nodes. Because normalization is functorial, we obtain $f^\nu \in \text{Aut}(C^\nu)$ and without loss of generality assume that $f^\nu(p_1) = p'_1$. We get an isomorphism

$$f_{p_1}^\nu : \mathbb{C}[[x']] \xrightarrow{\sim} \mathbb{C}[[x]].$$

Identify x with x' . The isomorphism can be characterized by $x \mapsto g(x)$ with the linear term of $g(x)$ nonzero (see [O'M70]). Say a is the linear term of g . Because C is stable, $\text{Aut}(C)$ is finite and thus there is a positive integer n such that $f^n = \text{id}_C$. This means that $(f^\nu)^n = \text{id}_{C^\nu}$ and therefore $a \in \mathbb{C}^\times$ is a root of unity. Define the coordinate

$$\bar{x} := \frac{1}{n} \sum_{i=0}^{n-1} \frac{1}{a^i} (f_{p_1}^\nu)^i(x).$$

It is clear that $f_{p_1}^\nu(\bar{x}) = ax$ and the linear term of $\bar{x}$ is $1 \in \mathbb{C}^\times$ meaning, it defines an isomorphism. Do exactly the same for $f_{p_2}^\nu(\bar{y}) = by$ with b also a root of unity. Because of the embeddings

$$\frac{\mathbb{C}[[x,y]]}{(xy)} \hookrightarrow \mathbb{C}[[x]] \oplus \mathbb{C}[[y]], \quad \frac{\mathbb{C}[[x',y']]}{(x'y')} \hookrightarrow \mathbb{C}[[x']] \oplus \mathbb{C}[[y']]$$

we find that after base changing to suitable coordinates,

$$f(\bar{x}) = ax, \quad f(\bar{y}) = by.$$

Notice that if $f(p_1) = p_2$ exactly the same would happen but we would switch the linear terms a and b . $\square$

Nodal curves have a rich combinatorial structure, captured by an object which is called the dual graph.

Definition 2.3 ([ACG⁺11, pp. 88]). *Let C be a connected nodal curve. Then we may associate a graph to it, called the dual graph $\Gamma(C)$, with for each irreducible component a vertex and for each node intersecting two irreducible components an edge between the two vertices corresponding to the irreducible components.*

2.1. Families of curves. We will start our study of families of curves varying over an analytic space.

Definition 2.4 ([ACG⁺11, Ch. 11.2]). *A family of genus g curves over an analytic space B is a flat proper map $f : \mathcal{C} \rightarrow B$ of finite-type such that each fiber $C_t := f^{-1}(t)$ with $t \in B$ is a connected genus g curve. We call $\mathcal{C}$ the total space and B the base of the family.*

Lemma 2.5. *Let $\mathcal{C} \rightarrow \Delta$ be a family of genus g curves with smooth general fiber and nodal special fiber C_0 . Then there is an open neighbourhood $U \subseteq \mathcal{C}$ with an integer $n \geq 1$ that allows an open immersion $U \hookrightarrow \{xy = t^n\} \subset \mathbb{C}^2 \times \Delta$ such that*

$$\begin{array}{ccccc} U & \hookrightarrow & \mathcal{C} & \longrightarrow & \Delta \\ \downarrow & & \nearrow \pi & & \\ \{xy = t^n\} & & & & \end{array}$$

commutes. Here $\pi : (x, y, t) \mapsto t$.

Proof. We aim to apply [Liu02, Lemma 10.3.20d] with $A = \mathcal{O}_{\Delta,0} = \mathbb{C}[[t]] \rightarrow \mathcal{O}_{\mathcal{C},e} = B$. Now

$$\begin{array}{ccccc} \mathbb{C}[[t]]/(t) & \longrightarrow & \mathcal{O}_{\mathcal{C},e} \otimes_{\mathbb{C}[[t]]} \mathbb{C}[[t]]/(t) & \xrightarrow{\sim} & \mathcal{O}_{\mathcal{C},e}/(t)\mathcal{O}_{\mathcal{C},e} \\ \uparrow & & \downarrow & & \\ \mathbb{C}[[t]] & \longrightarrow & \mathcal{O}_{\mathcal{C},e} & & \end{array}$$

But then since the fiber C_0 is defined as the pullback of $\text{Spec } \mathbb{C}[[t]]/(t) \rightarrow \Delta$ we get $\mathcal{O}_{\mathcal{C},e}/(t)\mathcal{O}_{\mathcal{C},e} = \mathcal{O}_{C_0,e}$. Also, since e is a node of the fiber C_0 , the local ring at e has the form $\widehat{\mathcal{O}_{C_0,e}} \cong \mathbb{C}[[x, y]]/(xy)$. Thus applying the Lemma gives us

$$\begin{aligned} \widehat{\mathcal{O}_{\mathcal{C},e}} &\cong \frac{\mathbb{C}[[t]][[x, y]]}{(xy - f(t))} \\ &\cong \frac{\mathbb{C}[[x, y, t]]}{(xy - f(t))} \end{aligned}$$

for some $f(t) \in (t)$. But then f can be written as

$$f(t) = t^n(a_n + a_{n+1}t + \dots)$$

where n is the first nonzero term which is greater than or equal to 1. Because a_n is nonzero, the power series $a_n + a_{n+1}t + \dots$ is invertible. This means that $f(t) = u(t)t^n$ with $u(t)$ invertible. Then the change of variables $x \mapsto u(t)x$ gives us

$$\widehat{\mathcal{O}_{\mathcal{C},e}} \cong \frac{\mathbb{C}[[x, y, t]]}{(xy - t^n)}.$$

From an analytic perspective, the node $e \in C_0$ thus has an open neighbourhood U that is isomorphic to an open in $\{xy = t^n\}$. $\square$

In particular, note that $\mathcal{C}$ is smooth if and only if $n = 1$. We can upgrade this lemma to a family over Δ^k for any positive k .

Lemma 2.6. *Suppose $\mathcal{C} \rightarrow \Delta^k$ is a family of nodal genus g curves with smooth fibers above $(\Delta^*)^k$. Denote by t_i the coordinates of Δ^k and let $I \subseteq [k]$ and $q \in \bigcap_{i \in I} \{t_i = 0\} \subseteq \Delta^k$ be a point. Suppose that $e \in \mathcal{C}_q$ is a node in the fiber above q . Then*

- (i) *there is an open neighbourhood of e , say $U \subseteq \mathcal{C}$ with integers $n_i \geq 0$ for each $i \in I$, such that*

$$\begin{array}{ccc} U & \xrightarrow{\quad} & \mathcal{C} \xrightarrow{\quad} \Delta^k \\ \downarrow & & \nearrow \pi \\ \{xy = \prod_{i \in I} t_i^{n_i}\} \times \prod_{i \notin I} \Delta & & \end{array}$$

commutes. Here the map π is similar as in Lemma 2.5 and one of the n_i satisfies $n_i \geq 1$.

- (ii) *The space $\mathcal{C}$ is smooth at e if and only if $\sum_{i \in I} n_i = 1$ (so only one $n_i = 1$ and the rest is 0).*
- (iii) *The space $\mathcal{C}$ is smooth at e if and only if there is a unique $j \in I$ and a point $q_j \in \{t_j = 0\} \setminus \bigcup_{i \in [k] \setminus \{j\}} \{t_i = 0\}$ such that e is a specialization of a node $e_i \in \mathcal{C}_{q_i}$.*

Proof. (i): Generalization of Lemma 2.5.

(ii): Clear.

(iii): Say there are two points e_1 and e_2 specializing to e . They both have local equations $\{xy = \prod_{i \in I} t_i^{n_i}\}$ and thus the local equation of e contains at least a factor $t_i t_j$ for $i, j \in I$ i.e. $\{xy = t_i t_j \prod_{k \in K} t_k\}$ for some collection $K \subseteq I$. $\square$

2.2. Deformation theory of curves. The main reference for this subsection is [ACG⁺11].

Definition 2.7 ([ACG⁺11, Ch. 11.2]). *Let $\varphi : \mathcal{C} \rightarrow B$ be a family of genus g curves with $\mathcal{C}$ smooth and $0 \in B$ a distinguished point. Suppose moreover that there is an isomorphism $\psi : \mathcal{C} \rightarrow \mathcal{C}_0$. Then we call the pair (φ, ψ) a deformation of $\mathcal{C}$ over B and such a deformation will often be denoted simply by $\mathcal{C} \rightarrow B$. A morphism of*

deformations $(f, F) : (\varphi, \psi) \rightarrow (\varphi', \psi')$ is a cartesian diagram

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{F} & \mathcal{C}' \\ \downarrow \varphi & & \downarrow \varphi' \\ B & \xrightarrow{f} & B' \end{array}$$

such that $f : 0 \mapsto 0$ and $F_0 = \psi' \circ \psi^{-1}$ as maps $\mathcal{C}_0 \rightarrow \mathcal{C}'_0$. We call a deformation $(\varphi : \mathcal{C} \rightarrow B, \psi : C \rightarrow \mathcal{C}_0)$ universal if for any other deformation $(\varphi : \mathcal{C} \rightarrow B, \psi : C \rightarrow \mathcal{C}_0)$ there is an open neighborhood $U \subseteq B'$ of 0 such that there is a map of deformations $(f, F) : (\varphi'_U, \psi') \rightarrow (\varphi, \psi)$ with $\varphi'_U : \mathcal{C}'_U := \mathcal{C}' \times_{B'} U \rightarrow U$ that induces a commutative diagram

$$\begin{array}{ccc} \mathcal{C}_0 & \xrightarrow{F_0} & \mathcal{C}'_0 \\ \psi^{-1} \downarrow & \nearrow \psi' & \\ C & & \end{array}$$

When a universal deformation space of a genus g curve C exists, we will denote it by $\text{Def}(C)$.¹

Deformations of curves always exist. We are particularly interested in the case of universal deformations which is neatly characterized.

Theorem 2.8 ([ACG⁺11, Thm. 11.3.17 and 11.4.3]). *Given C a nodal curve of genus g , there exists a universal deformation $\mathcal{C} \rightarrow \text{Def}(C)$ of C if and only if C is stable. If this is the case, then $\text{Def}(C) \cong \Delta^{3g-3}$ where $\Delta \subseteq \mathbb{C}$ is the open unit disk in the complex plane. Moreover one may choose coordinates such that each coordinate corresponds to a smoothening at most one of the nodes.*

When a curve is stable, its automorphism group is finite. The action of the automorphism group on the curve extends to compatible actions on the deformations of the curve. Indeed, if $a \in \text{Aut}(C)$ is an automorphism and $(\varphi : \mathcal{C} \rightarrow B, \psi : C \rightarrow \mathcal{C}_0)$ a universal deformation, $(\varphi, \psi \circ a)$ is also a universal deformation. Thus, up to possibly shrinking the base, we obtain a cartesian diagram

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{F_a} & \mathcal{C} \\ \downarrow & & \downarrow \\ B & \xrightarrow{f_a} & B \end{array}$$

where the horizontal arrows are isomorphisms and F_a restricted to 0 is the automorphism a . We may even assume that for two automorphisms a and a' , the induced automorphisms act functorially i.e., we have the equalities $F_{a \circ a'} = F_a \circ F_{a'}$ and $F_{\text{id}_C} = \text{id}_{\mathcal{C}}$ and similarly for f_a .

When we have an involution acting on the special fiber, we can calculate the involution induced on the base explicitly.

Lemma 2.9. *Say we have a stable curve C_0 with k nodes and an involution ι swapping two nodes e_i and e_j . Let $\mathcal{C} \rightarrow \text{Def}(C_0) = \Delta^{3g-3}$ be the universal deformation of C_0 and*

¹Universal deformations of curves are often called Kuranishy families in the literature.

denote by $t_1, \dots, t_{3g-3}$ its coordinates. Say t_i, t_j smoothen the nodes e_i, e_j respectively. Then, up to a coordinate change, the induced involution on $\text{Def}(C_0)$ maps $t_i \mapsto t_j$.

Proof. By Lemma 2.2, we may assume coordinates of e_i and e_j so that the involution acts linearly:

$$\begin{aligned} \iota : \{xy = 0\} &\rightarrow \{x'y' = 0\} \\ (x, y) &\mapsto (ax', by'), \end{aligned}$$

with $a, b \in \mathbb{C}^\times$ roots of unity. Note that we could also have $(x, y) \mapsto (ay', bx')$ but this does not matter as both involutions give the same map on the t parameter. Because ι is an involution $a, b = \pm 1$. Then the induced involution on $\Delta^{3g-2} \times \{xy = t\} \subseteq \mathcal{C}$ maps $t \mapsto abx'y' = \pm t'$, by continuity. In the minus case, perform a coordinate change $t' \mapsto -t'$. Then this means that the involution maps the fiber C_t to $C_{t'}$, and thus the involution on the base maps $t \mapsto t'$. $\square$

Lemma 2.10. *Say we have a stable curve C_0 with an involution ι fixing a node e . Let $\mathcal{C} \rightarrow \text{Def}(C_0) = \Delta^{3g-3}$ be the universal deformation of C_0 and t the smoothening parameter of e . Then for suitable coordinates, the induced involution on $\text{Def}(C_0)$ maps $t \mapsto t$ if the branches are swapped, and $t \mapsto \pm t$ if the branches are fixed. In the branchwise fixed case, $t \mapsto -t$ happens only when the involution acts as the identity on exactly one of the branches.*

Proof. We may assume again that the involution acts linearly on suitable coordinates $\{xy = 0\}$. We will consider the branch swapping case first. We may assume $\iota : (x, y) \mapsto \pm(y, x)$ because ι is an involution. Therefore, by continuity, ι acts on the smoothening parameter t in $\Delta^{3g-2} \times \{xy = t\} \subseteq \mathcal{C}$ as $\iota : t \mapsto t$.

Say branches are fixed. The involution will act as $(x, y) \mapsto (ax, by)$ where $a, b = \pm 1$, and if $a = 1$, $\iota : x \mapsto x$, meaning ι is the identity on that branch, and similarly for b . It follows that $\iota : t \mapsto abt = \pm t$. $\square$

Remark 2.11. From this discussion we see that if the involution on the special fiber has a finite number of fixed points, the induced involution on the base of the universal deformation swaps smoothening parameters of the nodes that are swapped, and acts trivially on the nodes that it fixes. In particular, a stable curve together with an involution (C_0, ι_0) having a finite amount of fixed points may be smoothened to (C_t, ι_t) where t is a general point in the fixed locus of the base B^ι . The families constructed this way will play a crucial role in the sequel.

2.3. Monodromy of curves. We recall the basics of monodromy. The main reference for this subsection is [Voi03, II.3].

Recall that if $f : X \rightarrow Y$ is continuous and $\mathcal{F}$ a sheaf on X , then $R^n f_* \mathcal{F}$ is the sheafification of the presheaf on Y defined by $U \mapsto H^n(f^{-1}(U), \mathcal{F})$. Therefore, if $\underline{\mathbb{Z}}$ is the locally constant sheaf with values in $\mathbb{Z}$ on the total space $\mathcal{C}$, the stalks of the higher

direct image sheaves can be computed as follows:

$$(R^n f_* \mathbb{Z})_t \cong H^n(C_t, \mathbb{Z}).$$

By Ehresmann's Theorem, for every t there is a small connected contractible neighbourhood U such that $f^{-1}(U) = C_t \times U$. Thus

$$\begin{aligned} (R^n f_* \mathbb{Z})(U) &:= H^n(f^{-1}(U), \mathbb{Z}) \\ &\cong H^n(C_t \times U, \mathbb{Z}) \\ &\cong H^n(C_t, \mathbb{Z}) \end{aligned}$$

and therefore $R^n f_* \mathbb{Z}$ is a local system with stalk $H^n(C_t, \mathbb{Z})$. From this we obtain the local system $(R^1 f_* \mathbb{Z})^\vee$ with stalk $H_1(C_t, \mathbb{Z})$. Following [Voi03, II.3.1], we obtain a *monodromy representation*

$$\pi_1(\Delta^*) \rightarrow \text{Aut}(H_1(C_t, \mathbb{Z})),$$

and we call the image of the generator $1 \in \pi_1(\Delta^*)$ the *monodromy operator* $T \in \text{Aut}(H_1(C_t, \mathbb{Z}))$.

2.4. Vanishing cycles. The following is an almost direct copy of [GFS25, Section 2.7], and is meant as a recollection on vanishing cycles.

Proposition 2.12. *Let $\varphi : \mathcal{C} \rightarrow \Delta$ be a family of genus g curves over the open disk $\Delta \subseteq \mathbb{C}$ with $\mathcal{C} \rightarrow \Delta$ smooth outside 0. Then $\mathcal{C}_0$ is a deformation retract of $\mathcal{C}$. This statement remains true if we replace Δ with any analytic space B up to shrinking B around 0.*

Proof. See [GFS25, Prop. 2.15]. □

Lemma 2.13. *Suppose $\mathcal{C} \rightarrow \Delta$ is a deformation of a genus g curve which is smooth outside 0. Suppose moreover that the special fiber $\mathcal{C}_0$ has as only singularities k distinct nodal points $\{e_i\}_{i=1}^k$. Then for C_t general there are k disjoint spheres $S_1^1, \dots, S_k^1 \subseteq C_t$ and a deformation retract of $\mathcal{C}$ to the union of C_t with k disjoint 2-dimensional balls $B_1^2, \dots, B_k^2$ glued along the pushout of*

$$\begin{array}{ccc} S_i^1 & \hookrightarrow & C_t \\ \downarrow & & \\ B_i^2 & & \end{array}$$

Proof. See [GFS25, Lemma 2.17]. □

In particular, the spheres S_i^1 define homology classes $\gamma_i \in H_1(C_t, \mathbb{Z})$ which we will call *vanishing cycles*.

Corollary 2.14. *For C_t a general fiber, the inclusion $C_t \hookrightarrow \mathcal{C}$ defines an exact sequence*

$$0 \longrightarrow \text{Van}_{C_t} \longrightarrow H_1(C_t, \mathbb{Z}) \xrightarrow{i_*} H_1(\mathcal{C}, \mathbb{Z}) \longrightarrow 0$$

where the module Van_{C_t} is generated by the vanishing cycles γ_i .

Proof. Follows from Proposition 2.12 and Lemma 2.13. $\square$

Proposition 2.15. [Picard-Lefschetz formula] *The monodromy operator $T \in \text{Aut}(H_1(C_t, \mathbb{Z}))$ can be calculated as*

$$T(x) = x - \sum_{\gamma} (x \cdot \gamma) \gamma$$

where $(- \cdot -)$ is the intersection pairing and γ runs over the vanishing cycles.

Proof. Generalization of [Voi03, Thm. 3.16]. See also [GFS25, Lemma 2.19]. $\square$

2.5. Monodromy over Higher Dimensional Bases. Suppose $\mathcal{C} \rightarrow \Delta^k$ is family of nodal genus g curves with smooth fibers above $(\Delta^*)^k$ and total space $\mathcal{C}$ smooth. We will construct vanishing cycles of this family by reducing to the one dimensional disk Δ .

Lemma 2.16. *Let $e \in [k]$. There is an open disk $B_e \cong \Delta$ that intersects $\{t_e = 0\}$ transversally at a point $q \in \{t_e = 0\}$ that does not lie on any other $\{t_i = 0\}$ for $i \neq e$.*

Proof. Take for example $q = (u_1, \dots, u_{e-1}, 0, u_{e+1}, \dots, u_k)$ and

$$B_e = \{(u_1, \dots, u_{e-1}, z, u_{e+1}, \dots, u_k) \mid z \in \Delta\},$$

which proves the lemma. $\square$

Lemma 2.17. *Let $\mathcal{C} \rightarrow \Delta^k$ be a family of genus g curves over Δ^k with smooth fibers above $(\Delta^*)^k$ and total space $\mathcal{C}$ smooth. Let B_e be a disk intersecting $\{t_e = 0\}$ transversally at a point q . Then the restriction $\mathcal{C}|_{B_e} := \mathcal{C} \times_{\Delta^k} B_e \rightarrow B_e \cong \Delta$ is a family of nodal genus g curves over Δ with total space smooth.*

Proof. First of all $B_e \cap \{t_e = 0\} = \{q\}$. Then the fibers on $B_e \setminus \{q\}$ are smooth, so we only need to check the fiber $(\mathcal{C} \times_{\Delta^k} B_e)_q \cong C_q$. So let $n \in C_q$ be a node. Then the local equation around n has the form $\{xy = t_e\} \times \prod_{i \neq e} \{t_i = 0\}$, thus

$$\begin{aligned} B_e \times_{\Delta^k} (\{xy = t_e\} \times \prod_{i \neq e} \Delta) &\cong B_e \times_{\Delta} \{xy = t_e\} \\ &\cong \{xy = z\} \end{aligned}$$

is the local equation for n in the restriction $\mathcal{C} \times_{\Delta^k} B_e \rightarrow B_e$ (the z comes from the definition of B_e). Applying Lemma 2.6 again gives us the result. $\square$

Let $f : \mathcal{C} \rightarrow \Delta^k$ be a family of nodal genus g curves over Δ^k with smooth fibers $(\Delta^*)^k$ and $\mathcal{C}$ smooth. Let $e \in C_0$ be a node in the special fiber which is a specialization of a node $e' \in C_q$ with $q \in \{t_e = 0\}$ as in Lemma 2.6.

Then using Lemma 2.17 the restriction $\mathcal{C}|_{B_e} \rightarrow B_e \cong \Delta$ is a family of nodal genus g curves over Δ smooth outside $q \in B_e$ and $(\mathcal{C}|_{B_e})_q$ a nodal curve. Using Lemma 2.13, the node $e' \in (\mathcal{C}|_{B_e})_q$ gives us a vanishing cycle

$$\gamma_{e'} \in H_1((\mathcal{C}|_{B_e})_{t'}, \mathbb{Z}) = H_1(C_{t'}, \mathbb{Z})$$

for $t' \in B_e \setminus \{q\}$.

Then for $t \in (\Delta^*)^k$, let $\xi : [0, 1] \rightarrow \Delta^k$ be a path from t to t' . Because $(R^1 f_* \mathbb{Z})^\vee$ is a local system with stalk $H_1(C_t, \mathbb{Z})$, after pulling back, the sheaf $\xi^{-1}(R^1 f_* \mathbb{Z})^\vee$ is also a local system with stalk $H_1(C_t, \mathbb{Z})$ on $[0, 1]$. Because $[0, 1]$ is simply connected, the sheaf $\xi^{-1}(R^1 f_* \mathbb{Z})^\vee$ is actually isomorphic to the constant sheaf $H_1(C_t, \mathbb{Z})$ on $[0, 1]$. Choose such an isomorphism. Then

$$H_1(C_t, \mathbb{Z}) \cong \xi^{-1}(R^1 f_* \mathbb{Z})_1^\vee \cong (R^1 f_* \mathbb{Z})_{t'}^\vee \cong H_1(C_{t'}, \mathbb{Z}).$$

Then we may transport the vanishing cycle $\gamma_{e'}$ to a vanishing cycle γ_e along these isomorphisms.

Definition 2.18. Let $f : \mathcal{C} \rightarrow \Delta^k$ be a family of nodal genus g curves over Δ^k with smooth fibers above $(\Delta^*)^k$ and $\mathcal{C}$ smooth. Let $e \in [k]$ and $t \in (\Delta^*)^k$. We define the monodromy operator T_e about the e 'th coordinate hyperplane via the diagram

$$\begin{array}{ccc} H_1(C_t, \mathbb{Z}) & \xrightarrow{\quad T_e \quad} & H_1(C_t, \mathbb{Z}) \\ \downarrow \wr & & \downarrow \wr \\ H_1(C_{t'_e}, \mathbb{Z}) & \xrightarrow{\quad T_e \quad} & H_1(C_{t'_e}, \mathbb{Z}) \end{array}$$

where $t'_e \in B_e \setminus \{q_e\}$ as in the previous discussion. Define the logarithm of the monodromy operator as $N_e = T_e - \text{id}$.

Proposition 2.19. Let $f : \mathcal{C} \rightarrow \Delta^k$ be a family of nodal genus g curves over Δ^k , with smooth fibers above $(\Delta^*)^k$ and $\mathcal{C}$ smooth. Let $l \leq k$ and suppose we have a map $g : \Delta^l \rightarrow \Delta^k$. Consider the induced family defined on the pullback

$$\begin{array}{ccc} \mathcal{C}' & \xrightarrow{G} & \mathcal{C} \\ f' \downarrow & & \downarrow f \\ \Delta^l & \xrightarrow{g} & \Delta^k \end{array}$$

Then labeling the monodromy representations ρ_F and ρ_f we obtain a commutative diagram

$$\begin{array}{ccc} \pi_1((\Delta^*)^l, t) & \xrightarrow{\rho_{f'}} & \text{Aut}(H_1(C_t, \mathbb{Z})) \\ g_* \downarrow & & \uparrow \wr \\ \pi_1((\Delta^*)^k, g(t)) & \xrightarrow{\rho_f} & \text{Aut}(H_1(C'_{g(t)}, \mathbb{Z})) \end{array}$$

Proof. Clear once tracing through the definitions of the monodromy representations. $\square$

3. ABELIAN VARIETIES

The following is a summary of abelian varieties. Everything stated here can be found in [BL13].

Let $X = V/\Lambda$ be a complex torus of complex dimension g , which consists of a complex

vector space V of dimension g and a lattice $\Lambda \subset V$. We have the exponential sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_X^* \longrightarrow 0$$

and an identification $H^1(X, \mathcal{O}_X^*) \cong \text{Pic } X$ providing us with a map $c_1 : \text{Pic } X \rightarrow H^2(X, \mathbb{Z})$ called the *first Chern class map*. The image of this map is called the *Néron-Severi group* denoted by $\text{NS}(X)$. On complex tori, this group can be characterized by the Hermitian forms on V such that the imaginary parts are integer-valued on the lattice Λ :

$$\text{NS } X \cong \{H : V \times V \rightarrow \mathbb{C} \text{ Hermitian form} \mid \text{Im } H(\Lambda, \Lambda) \subseteq \mathbb{Z}\}.$$

The imaginary part $\text{Im } H$ of such a Hermitian form H is an alternating form on V and we may associate a basis $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$ to Λ such that $\text{Im } H$ takes the shape

$$(2) \quad \begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix},$$

where $D = \text{diag}(d_1, \dots, d_g)$ for integers $d_i \geq 0$. The *elementary divisors* $d_1, \dots, d_g$ are uniquely determined by $c_1(\mathcal{L}) = L \in \text{NS } X$ and thus by $\mathcal{L} \in \text{Pic } X$. The vector $(d_1, \dots, d_g)$ is called the *type* of a line bundle $\mathcal{L}$ on X .

A *polarization* on a complex torus $X = V/\Lambda$ is the first Chern class of an ample line bundle $\mathcal{L}$ on X , or equivalently, the first chern class of a line bundle such that all its elementary divisors are positive. A pair (X, L) with X a complex torus and $L = c_1(\mathcal{L})$ a polarization is called an *abelian variety*, and if $\mathcal{L}$ has type $(1, \dots, 1)$ it is called a *principally polarized abelian variety*.

Example 3.1 ([BL13, Ch. 11]). If C is a smooth projective genus g complex curve, it determines a principally polarized abelian variety: the first homology $H_1(C, \mathbb{Z})$ defines a lattice in $H^0(C, \omega_C)^\vee$ via Stokes' theorem, where $H^0(C, \omega_C)$ is the vector space of holomorphic 1-forms, and define the *Jacobian (variety) of C* by

$$J(C) := \frac{H^0(C, \omega_C)^\vee}{H_1(C, \mathbb{Z})}.$$

This is a complex torus of dimension g . The Jacobian variety admits a canonical principal polarization making it into a principally polarized abelian variety. Any divisor Θ on $J(C)$ such that the line bundle $\mathcal{O}_{J(C)}(\Theta)$ defines the canonical divisor is called a *theta divisor* of the Jacobian $J(C)$. The theta divisor of the Jacobian can be identified with the intersection pairing on the integral homology of the curve and we will freely do so in the sequel.

Example 3.2 ([BL13, Ch. 12]). Other examples coming from curves are Prym varieties. Given an étale² double cover $f : C \rightarrow C'$ of smooth projective curves of genera g

²If the double cover has two ramification points, it still defines a principally polarized abelian variety. This will however not define points of $\mathcal{R}_g$, the moduli space of étale double covers of genus g curves, which we restrict our attention to.

and g' , consider the *norm map*:

$$\begin{aligned} \mathrm{Nm}_f : J(C) &\rightarrow J(C') \\ \mathcal{O}_C(\sum_i r_i p_i) &\mapsto \mathcal{O}_{C'}(\sum_i r_i f(p_i)). \end{aligned}$$

Then the *Prym variety* associated to the étale double cover f is defined to be the connected component containing $0 \in J(C)$ of the kernel of the norm map Nm_f :

$$P(f) := \ker(\mathrm{Nm}_f)_0 \subseteq J(C).$$

It is an abelian subvariety of $J(C)$ and obtains a principal polarization which we denote by Ξ . The theta divisors of $J(C)$ and $P(f)$ are related by the identity

$$2\Xi = \Theta|_{P(f)}.$$

Because $f : C \rightarrow C'$ is a double cover, it induces an involution $\iota : C \rightarrow C$ and C' can be identified with the quotient C/ι . The involution ι induces actions on $H^0(C, \omega_C)$ and $H_1(C, \mathbb{Z})$ and it can be shown (see [BL13, Prop. 12.4.2.]) that

$$(3) \quad P(f) = \frac{(H^0(C, \omega_C)^-)^{\vee}}{H_1(C, \mathbb{Z})^-}$$

where $H^0(C, \omega_C)^-$ and $H_1(C, \mathbb{Z})^-$ mean the (-1) -eigenspaces of the modules. This characterization of the Prym variety will be used throughout the rest of this paper.

3.1. The Monodromy Filtration for Abelian Varieties. A reference for this section is [EGFS25a, Section 2.3]. Throughout this section, let $f : X \rightarrow \Delta^k$ be a family of projective varieties such that general fibers are principally polarized abelian varieties. We call such families *degenerations*. Suppose that, for such a degeneration, the general fibers over $\{t_i = 0\}$ have simple normal crossings. Let X_t be a general fiber. Similarly to Section 2.3, we obtain monodromy operators $T_i : H_1(X_t, \mathbb{Z}) \rightarrow H_1(X_t, \mathbb{Z})$ about the i 'th coordinate hyperplane. Since the general fibers over $\{t_i = 0\}$ have simple normal crossings, by [Cle69, Thm. 7.36], the monodromy operator $T = \prod_i T_i$ is unipotent (the product here is composition). Hence $N = T - \mathrm{id}$ is nilpotent and in fact, $N^2 = 0$. Then by [Sch73, Lemma 6.4] the monodromy weight filtration only consists of

$$0 \subset W_{-2} \subset W_{-1} \subset W_0 = H_1(X_t, \mathbb{Z}).$$

In this case, the weight filtration has specific description $W_{-2} = (\mathrm{im} N)^{\mathrm{sat}}$ and $W_{-1} = \ker N$ (see [EGFS25a, Section 2.3]). For future reference, we will state some well-known linear algebra results on the monodromy filtration.

Lemma 3.3. *We have an isomorphism $W_{-2} \cong (\mathrm{gr}_0^W)^*$.*

Proof. Because X_t carries a principal polarization, say L_t , it defines a unimodular form on W_0 giving us a perfect pairing

$$\begin{aligned} W_0 \times W_0 &\rightarrow \mathbb{Z} \\ (x, y) &\mapsto L_t(x, y). \end{aligned}$$

Now this restricts to

$$(4) \quad W_{-2} \times \frac{W_0}{W_{-1}} \rightarrow \mathbb{Z}$$

because if $(x, y - y') \in W_{-2} \times W_0$, with $y - y' \in W_{-1}$, then

$$(x, y - y') \mapsto L_t(x, (y - y')) = 0,$$

because $W_{-1} \cong W_{-2}^\perp$ [Sch73, Lemma 6.4]. Thus the map (4) is well defined. The injectivity of the resulting map $W_{-2} \rightarrow \left(\frac{W_0}{W_{-1}}\right)^\vee$ follows, because if $x \in W_{-2}$ such that $L_t(-, x) = 0$, then nondegeneracy implies $x = 0$. Now suppose $\bar{l} : \frac{W_0}{W_{-1}} \rightarrow \mathbb{Z}$ is a linear map. This lifts to a linear map $l : W_0 \rightarrow \mathbb{Z}$ which we know by perfectness is equal to $L_t(-, x)$ for some $x \in W_0$. Moreover, we have the commuting diagram

$$\begin{array}{ccccc} W_{-1} & \hookrightarrow & W_0 & \longrightarrow & \frac{W_0}{W_{-1}} \\ & & \downarrow l & & \downarrow \bar{l} \\ & & \mathbb{Z} & \xrightarrow{=} & \mathbb{Z} \end{array}$$

with the top row exact, meaning that $L_t(-, x)|_{W_{-1}} = 0$. Thus for every $y \in W_{-1}$, we have that $l(y) = L_t(y, x) = 0$ which means that $x \in W_{-1}^\perp = W_{-2}$. So $L_t(-, x) \mapsto \bar{l}$ meaning the map $W_{-2} \rightarrow \frac{W_0}{W_{-1}}$ is surjective and thus, this map is an isomorphism. $\square$

Lemma 3.4. *Let V be a free $\mathbb{Z}$ -module and ι an involution on V . Assume V admits a bilinear, nondegenerate form L such that $L(\iota(x), \iota(y)) = L(x, y)$ for each $x, y \in V$. Denote by L^- the restriction of L to V^- . Then for any $W \subseteq V$ a submodule, $(W^-)^{\perp_{L^-}} = (W^{\perp_L})^-$ as submodules of V^- .*

Proof. Because $(W^{\perp_L})^- = V^- \cap W^{\perp_L}$, the reverse inclusion is clear.

Let $v \in (W^-)^{\perp_{L^-}}$ and $w \in W$. Then $L(v, \iota(w)) = -L(v, w)$, meaning $0 = L(v, w - \iota(w)) = 2L(v, w)$ and thus $L(v, w) = 0$. $\square$

4. MATROIDAL FAMILIES

Matroids are combinatorial devices that track independence, similar to linear independence in vector spaces. We will first recall some of the basics of matroid theory.

Definition 4.1 ([EGFS25a, Def. 4.1]). *A matroid is a pair $(E, \mathcal{I})$ where E is a finite set and $\mathcal{I}$ is a collection of subsets of E , called the independent sets, such that:*

- (i) *the empty set is independent;*
- (ii) *any subset of an independent set is independent;*
- (iii) *if I and J are independent sets with $|I| > |J|$ then there is an element $i \in I \setminus J$ such that $J \cup \{i\}$ is independent.*

The finite set E is called the ground set and often omitted from the notation. A basis is a maximal independent subset of the ground set and all bases have the same cardinality. The cardinality of such a basis is called the rank of the matroid. An isomorphism of

matroids is a set theoretic map on the ground sets such that a set is independent if and only if its image is.

Example 4.2. If $\underline{R}$ is a matroid, we can define the *dual matroid* $\underline{R}^*$ to be the matroid on the same ground set as $\underline{R}$ whose bases³ are the complements of the bases of $\underline{R}$.

Example 4.3. Let K be a field and φ_K a matrix with values in K . The matrix defines a matroid $\text{col}(\varphi_K)$ on the ground set $E := \{\text{columns of } \varphi_K\}$ whose independent sets are precisely the ones such that the set of column vectors is linearly independent over K . Any matroid isomorphic to the column matroid of a matrix over K is called *K-representable* and we call φ_K a *K-realization of the matroid*.

Matroids that are *K-representable* for every field K form an important class of matroids. They are called *regular matroids*. We call an integer matrix φ an *integral realization* of $\underline{R}$ if φ_K realizes the matroid $\underline{R}$ for all fields K (see [EGFS25a, Def. 4.2]). Regular matroids are neatly characterized by work of Tutte (see [Tut58a] and [Tut58b]):

Theorem 4.4. *A matroid is regular if and only if it can be represented by a totally unimodular matrix, which is a matrix such that all of its minors have determinant ± 1 or 0.*

Remark 4.5. Notice that an integral realization need not be totally unimodular. Indeed, any matrix of the form $U \cdot T$ with U a square unimodular matrix and T a totally unimodular matrix is an integral realization of a regular matroid (apply the results in [Oxl06, section 6.3]).

We will encounter matroids which are representable over all fields, except possibly over fields of characteristic two, which are called *near-regular matroids* (see [Oxl06, pp. 557]). For this, we introduce the following definition:

Definition 4.6. *An integer valued matrix is called a near-integral realization if it realizes a near-regular matroid over all fields of characteristic not equal two.*

Example 4.7. Let G be a graph with $E(G)$ its set of edges. When we choose an orientation of the edges, we have an inclusion

$$H_1(G, \mathbb{Z}) \hookrightarrow \mathbb{Z}^E.$$

Consider the canonical basis of $\mathbb{Z}^{E(G)}$, together with the coordinate functions $\tilde{x}_e : \mathbb{Z}^{E(G)} \rightarrow \mathbb{Z}$. Restrict to first homology to get maps $x_e := \tilde{x}_e|_{H_1(G, \mathbb{Z})} : H_1(G, \mathbb{Z}) \rightarrow \mathbb{Z}$ giving us

$$(5) \quad \begin{aligned} E(G) &\rightarrow H_1(G, \mathbb{Z})^\vee, \\ e &\mapsto x_e. \end{aligned}$$

The *cographic matroid* of G , denoted by $M^*(G)$, is defined to be the matroid on the realization (5) over $\mathbb{R}$. Their duals are called *graphic matroids*, denoted by $M(G)$ and

³Notice that the bases of a matroid encode the entire structure of the matroid: a subset is independent if and only if it is contained in a basis.

it is a well-known theorem that a cographic matroid is graphic if and only if their underlying graphs are planar graphs (see [Oxl06, Thm 5.2.2]).

Remark 4.8. Graphic and cographic matroids are always regular. Their regularity can be proven by analyzing the incidence matrix of the graph (see [P⁺14, Section 4.1], of which we will use a generalizations in later chapters. The realization (5) is thus an integral realization.

Example 4.9. A famous example of a regular matroid is R_{10} , discovered by [Sey80]. It is the matroid on the column vectors of the totally unimodular matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & -1 \end{pmatrix}$$

and is neither graphic nor cographic. It is shown that every regular matroid can be obtained by so-called 1-sums, 2-sums and 3-sums of graphic and cographic matroids, and copies of R_{10} (see [Oxl06, Thm. 13.1.1.]).

Definition 4.10 ([EGFS25a, Def. 2.6.]). *Let $X^* \rightarrow (\Delta^*)^k$ be a degeneration of principally polarized abelian varieties such that the monodromy is unipotent. Let $e \in [k]$ and denote by L_t the principal polarization of a general fiber X_t . The monodromy bilinear form on $U := \mathrm{gr}_0^W H_1(X_t, \mathbb{Z})$ about the e 'th coordinate hyperplane is defined by*

$$B_e : (x, y) \mapsto L_t(N_e(x), y).$$

We define the monodromy cone to be

$$\mathbb{R}_{\geq 0} \{B_e \mid e \in [k]\} \subseteq U \otimes_{\mathbb{Z}} \mathbb{R}$$

and we say it is a matroidal cone if $B_e = d_e y_e^2$ for some postive integers d_e and linear forms y_e on $\mathrm{gr}_0^W H_1(X_t, \mathbb{Z})$ such that the map $e \mapsto y_e$ defines an integral realization of a regular matroid $\underline{R}$.

We now get to the central definition of [EGFS25b].

Definition 4.11 ([EGFS25b, Def. 4.1.]). *Let $\pi : X^* \rightarrow (\Delta^*)^k$ be a smooth projective family of g -dimensional principally polarized abelian varieties. We say that π^* is a matroidal family associated to a regular matroid $\underline{R}$ with integral realization $E \rightarrow U^\vee$, $e \mapsto y_e$ for some free $\mathbb{Z}$ -module U of rank $g = \mathrm{rank}(\underline{R})$, if the following holds, with $t \in (\Delta^*)^k$:*

- (1) *the monodromy of the family is unipotent;*
- (2) *we have an isomorphism $U \cong \mathrm{gr}_0^W H_1(X_t, \mathbb{Z})$;*
- (3) *for each $e \in E$, there is a positive integer d_e , such that the bilinear form $d_e y_e^2$ on U agrees with the monodromy bilinear form B_e on $\mathrm{gr}_0^W H_1(X_t, \mathbb{Z})$.*

Observe that a degeneration is matroidal if and only if it is *maximally degenerate*, meaning $W_{-2} = W_{-1}$, and the monodromy cone is a matroidal cone.

Similarly, we will call a family $X^* \rightarrow (\Delta^*)^k$ of principally polarized abelian varieties *near-matroidal* if the forms $e \mapsto y_e$ form a near-integral realization of a near regular matroid

4.1. Matroidal families of Jacobians. Let $\mathcal{J}^* \rightarrow (\Delta^*)^k$ be a family of Jacobians arising from a degeneration of curves $\mathcal{C} \rightarrow \Delta^k$ with smooth total space. The goal of this subsection is to show that the monodromy cone of the family $\mathcal{J}^* \rightarrow (\Delta^*)^k$ is matroidal with cographic matroid $M^*(\Gamma(C_0))$. The results in this section are well known (see for example [EGFS25a, Ex. 2.15.]). We have a canonical isomorphism $H_1(J(C_t), \mathbb{Z}) \cong H_1(C_t, \mathbb{Z})$ and the monodromy operators agree. It thus suffices to reason over curves. We will denote by $E(\Gamma(C_0))$ the set of nodes of the central fiber of the degeneration and we will freely identify the theta divisor of Jacobians with the intersection product on the corresponding curve.

Lemma 4.12. *For each $e \in E(\Gamma(C_0))$, we can find a cycle $\alpha_e \in H_1(C_t, \mathbb{Z})$ such that $\Theta_t(\alpha_e, \gamma_e) = 1$ and $\Theta_t(\alpha_e, \gamma_u) = 0$ for $u \neq e$.*

Proof. Recall from Corollary 2.14 that we have an exact sequence

$$0 \longrightarrow \text{Van}_{C_t} \longrightarrow H_1(C_t, \mathbb{Z}) \longrightarrow H_1(\mathcal{C}, \mathbb{Z}) \longrightarrow 0.$$

Since $H_1(\mathcal{C}, \mathbb{Z})$ is torsion free, the sequence is split. Define for each $e \in E(\Gamma(C_0))$ the map

$$f_e : H_1(C_t, \mathbb{Z}) \cong \text{Van}_{C_t} \oplus H_1(\mathcal{C}, \mathbb{Z}) \rightarrow \mathbb{Z}$$

to be such that only $f_e : \gamma_e \mapsto 1$, all other $\gamma_u \mapsto 0$ different from e and $f_e : x \mapsto 0$ for each $x \in H_1(\mathcal{C}, \mathbb{Z})$. The intersection pairing Θ_t is unimodular and we may thus identify each homology class $x \in H_1(C_t, \mathbb{Z})$ with $\Theta_t(-, x) \in H_1(C_t, \mathbb{Z})^\vee$. Now $f_e \in H_1(C_t, \mathbb{Z})^\vee$ and thus there is an $\alpha_e \in H_1(C_t, \mathbb{Z})$ such that $f_e = \Theta_t(-, \alpha_e)$, meaning that $1 = f_e(\gamma_e) = \Theta_t(\gamma_e, \alpha_e)$. Moreover, for $u \neq e$ we have $0 = f_e(\gamma_u) = \Theta_t(\gamma_u, \alpha_e)$. $\square$

Definition 4.13. *Let $\mathcal{C} \rightarrow \Delta^k$ be a family of nodal curves. We say that a node e in the central fiber C_0 has color i if it is the specialization of a node in some fiber above $\{t_i = 0\}$. As remarked in property (iv) of Lemma 2.6, every node e has a specialization to a node e_i in a fiber over a unique hyperplane $\{t_i = 0\}$. This thus defines a coloring of the edges $E(\Gamma(C_0))$ of the dual graph of the central fiber:*

$$E(\Gamma(C_0)) = \coprod_i E_i.$$

Lemma 4.14. *The logarithm of the i -monodromy has the form*

$$N_i(x) = - \sum_{e \in E_i} \Theta_t(x, \gamma_e) \gamma_e$$

Proof. Apply the Picard-Lefschetz formula (see Prop. 2.15) and use the identity $N_i = T_i - \text{id}$. $\square$

Corollary 4.15. *We have the equality $W_{-2} = \text{Van}_{C_t}$*

Proof. Recall $W_{-2} := (\text{im } N)^{\text{sat}}$ and apply Lemma 4.12 together with Lemma 4.14 $\square$

It follows that the monodromy bilinear forms B_i can be computed as

$$B_i(a, b) = \sum_{e \in E_i} \Theta_t(a, \gamma_e) \Theta_t(b, \gamma_e).$$

Proposition 4.16. *We have a canonical isomorphism*

$$\mathrm{gr}_0^W H_1(C_t, \mathbb{Z}) \cong H_1(\Gamma(C_0), \mathbb{Z})$$

Moreover, after choosing an orientation of the dual graph of the central fiber⁴, the diagram

$$\begin{array}{ccc} \mathbb{Z}^{E(\Gamma(C_0))} & \xrightarrow{x_e} & \mathbb{Z} \\ \uparrow & \searrow & \\ H_1(\Gamma(C_0), \mathbb{Z}) & & \end{array}$$

commutes with $x_e : \mathbb{Z}^{E(\Gamma(C_0))} \rightarrow \mathbb{Z}$ the coordinate projection and $H_1(\Gamma(C_0), \mathbb{Z}) \rightarrow \mathbb{Z}$ the linear map corresponding to $x \mapsto \Theta_t(x, \gamma_e)$ along the isomorphism.

Proof. See [EGFS25a, Prop. 5.12]. □

In conclusion

$$(6) \quad B_i(a, b) = \sum_{e \in E_i} x_e(a) \cdot x_e(b),$$

for $a, b \in \mathrm{gr}_0^W H_1(C_t, \mathbb{Z})$. We restrict now our attention to the case when the central fiber is stable.

Proposition 4.17. *Let C_0 be a stable curve with $k \leq 3g-3$ nodes and $\mathcal{C} \rightarrow \Delta^k \subseteq \Delta^{3g-3}$ the slice of its universal deformation where the nodes of C_0 are smoothed according to their respective directions. Then the monodromy cone associated to the degeneration $\mathcal{J}\mathcal{C}^* \rightarrow (\Delta^*)^k$ is matroidal with matroid $M^*(\Gamma(C_0))$.*

Proof. This follows from the isomorphisms $W_{-2}H_1(C_t, \mathbb{Z}) \cong \mathrm{gr}_0^W H_1(C_t, \mathbb{Z})^\vee$ in Lemma 3.3 and $\mathrm{gr}_0^W H_1(C_t, \mathbb{Z}) \cong H_1(\Gamma(C_0), \mathbb{Z})$ of Proposition 4.16. That the monodromy bilinear forms have the shape $d_e x_e^2$ follows from equation (6) and the fact that in this slice of the universal deformation, there is exactly one node over each $\{t_i = 0\}$. Thus $B_e = x_e^2$. We conclude by the fact that cographic matroids are regular and that the forms x_e realize $M^*(\Gamma(C_0))$. □

5. MONODROMY OF PRYM VARIETIES

We will calculate the monodromy of families of Prym varieties in this chapter. Let us first start with defining the families of Prym varieties we will be analyzing throughout this master's thesis.

⁴Choosing an orientation of the dual graph of the central fiber corresponds to choosing an orientation of the vanishing cycles in the homology of a smooth fiber.

5.1. Families of Prym varieties. Say C_0 is a stable curve with an involution ι_0 . Recall from Section 2.2 that the involution extends to the universal deformation

$$\mathcal{C} \rightarrow \text{Def}(C_0) = \Delta^{3g-3}$$

of C_0 compatibly, giving us involutions $f_t : \text{Def}(C_0) \rightarrow \text{Def}(C_0)$ and $F_t : \mathcal{C} \rightarrow \mathcal{C}$. We obtain a family of smooth curves together with an involution $(\mathcal{C}, F_t) \rightarrow (\Delta^{3g-3})^{f_t}$ so that the general fiber (C_t, ι_t) has involution $\iota_t := F_t|_{C_t}$. We want to deduce when this will induce a degeneration of Prym varieties. For this, we need to examine the extra fixed points obtained in a nearby smooth fiber (C_t, ι_t) . First, classify the nodes:

Definition 5.1 ([ABH01, pp. 6]). *Let C_0 be a nodal curve, e a node and ι an involution on C_0 . We call e a*

- (1) *branchwise fixed node if $\iota(e) = e$ and locally fixes the branches connected by e ;*
- (2) *swapping fixed node if $\iota(e) = e$ and locally swaps the branches connected by e ;*
- (3) *non-fixed node if $\iota(e) \neq e$.*

Let e be a node in C_0 and γ the vanishing cycle in C_t corresponding to it. We may view γ as a submanifold of C_t by definition (see Lemma 2.13). The extra fixed points obtained in a nearby smoothening of a fiber must necessarily lie on the vanishing cycles associated to fixed nodes. The following lemma shows which nodes we want to exclude from our analysis. It seems that this lemma is known to the experts, but a proof remains absent in the literature.

Lemma 5.2. *Suppose we have a stable curve C_0 with an involution ι_0 having finitely many fixed points and let e fixed node. Let γ the vanishing cycle corresponding to the node in a nearby smooth fiber. Then if e is a swapping fixed node, the restriction of the involution ι_t to the submanifold γ has two fixed points. If e is a branchwise fixed node, the restriction of the involution has no fixed points.*

Proof. Let e be a swapping fixed node. Let $U := \{xy = t\}$ be the local equation of the universal deformation of the node and U_t the corresponding local equation of a nearby fiber. Because ι_0 swaps the irreducible components, $\{x = 0\}$ is swapped with $\{y = 0\}$. Using Lemma 2.2, we may assume the involution acts linearly meaning that on C_0 the involution has the form $(x, y) \mapsto (\pm y, \pm x)$, and thus by continuity, also on the nearby fiber U_t . The vanishing cycle γ has the form

$$\gamma = \{(x, y) \in U_t \mid |x| = |y|\},$$

(see [ACG⁺11, p. 147]) and from this equation it follows that $|x| = \sqrt{|t|}$. Thus

$$\begin{aligned} x &= \sqrt{|t|} e^{i\theta}, \\ y &= \sqrt{|t|} e^{i\varphi} \end{aligned}$$

and $\varphi = \theta_t - \theta$ where θ_t is the angle of t . Because the involution acts as $(x, y) \mapsto (\pm y, \pm x)$, on U_t , it sends the angle $\theta \mapsto \theta_t - \theta$ in the plus case, and $\theta \mapsto \theta_t - \theta + \pi$ in the minus case. These are both reflections along the axes $\frac{1}{2}\theta_t$ and $\frac{1}{2}(\theta_t + \pi)$, which

have exactly two fixed points. The case of a branchwise fixed node is similar: ι_0 cannot act trivially on the branches because we have only finitely many fixed points and conclude that the involution acts locally as $(x, y) \mapsto (-x, -y)$, which is a rotation on the vanishing cycle. The involution of C_t restricted to γ thus has no fixed points. $\square$

This means that only swapping fixed nodes contribute to additional fixed points on a smooth fiber.

Definition 5.3 ([Bea77]). *Let $C_0 \rightarrow C'_0$ be a double cover of stable curves. We say it is admissible if it has no smooth fixed points and no swapping fixed nodes. We say the induced involution on C_0 is an admissible involution if it induces an admissible double cover of stable curves. We say a double cover of graphs $\Gamma \rightarrow \Gamma'$ is admissible if the induced involution on Γ does not swap the vertices incident to a fixed edge. We define an admissible involution to be one such that its induced double cover will be admissible.*

Let C be a smooth genus g curve and ι a free involution on C . This gives an étale double cover $C \rightarrow C/\iota$. The datum of an étale double cover of a smooth curves is equivalent to defining a point $[C', \eta] \in \mathcal{R}_{g'}$ with

$$\mathcal{R}_{g'} := \{[C', \eta] \mid C' \text{ smooth curve of genus } g', \text{ and } \eta \in H^1(C', \mathbb{Z}/2\mathbb{Z}) \setminus \{0\}\}.$$

The forgetful map $\mathcal{R}_{g'} \rightarrow \mathcal{M}_{g'}$ is finite of degree $|H^1(C', \mathbb{Z}/2\mathbb{Z}) \setminus \{0\}| = 2^{2g'} - 1$, and thus

$$\dim(\mathcal{R}_{g'}) = \dim(\mathcal{M}_{g'}) = 3g' - 3.$$

Let $\overline{\mathcal{R}_{g'}}, \overline{\mathcal{M}_{g'}}$ denote the closures of $\mathcal{R}_{g'}, \mathcal{M}_{g'}$ respectively. The moduli space $\overline{\mathcal{R}_{g'}}$ is precisely the moduli space of admissible double covers of stable genus g curves (see and [Bea77]) and $\overline{\mathcal{M}_{g'}}$ is the moduli space of stable curves (see [DM69]). Then if $x = [C'_0, \eta] \in \overline{\mathcal{R}_{g'}}$, it defines an admissible double cover $C_0 \rightarrow C'_0$ and we have

$$\begin{aligned} \text{Def}(C_0, \iota_0) &= \text{Def}(C'_0, \eta) \\ &= T_x \overline{\mathcal{R}_{g'}} \\ &= \Delta^{3g'-3} \\ &= (\Delta^{3g-3})^{f_\iota} \\ &= \Delta^{k-j} \times \Delta^{3g'-3-k+j} \end{aligned}$$

where k is the amount of nodes in C_0 , j the pairs of exchanged nodes and f_ι the induced involution on $\text{Def}(C_0)$, which was calculated in Lemma 2.9 and 2.10.

Remark 5.4. Using Lemma 5.2, we conclude that every admissible double cover of stable curves with k nodes and j pairs of non-fixed nodes in C_0 , induces a degeneration of Prym varieties $P^* \rightarrow (\Delta^*)^{k-j} \times \Delta^{3g'-3-k+j}$.

These will be the families that we will analyze for the remainder of the thesis.

5.2. The monodromy weight filtration of degenerations of Prym varieties.

Let $C_0 \rightarrow C'_0$ be an admissible double cover of stable curves and say that C_0 has

$k \leq 3g - 3$ nodes. Denote by

$$e_1, \dots, e_{2j}, \\ e_{2j+1}, \dots, e_k$$

the nodes of C_0 such that the first $2j$ nodes are exchanged by ι_0 and others remain fixed. Notice that for $i \leq j$ there persist exactly two nodes, which are swapped under the involution, on hyperplanes $\{t_i = 0\}$ and for $i > j$ there persists exactly one node which remains fixed under the involution. Recall that we have a family of curves with an involution $(\mathcal{C}, F_\iota) \rightarrow \Delta^{k-j} \times \Delta^{3g'-3-k+j}$ inducing a degeneration of Prym varieties $P^* \rightarrow (\Delta^*)^{k-j} \times \Delta^{3g'-3-k+j}$. We will explicitly calculate the monodromy weight filtration for this degeneration.

Lemma 5.5. *The monodromy operators T_i^p on the family $(\mathcal{C}|_{\Delta^{k-j}}, F_\iota) \rightarrow \Delta^{k-j} \subseteq \text{Def}(C_0, \iota_0)$ of the slice of the universal deformation where the monodromy is nontrivial have the shape*

$$T_i^p = \begin{cases} T_{2i-1} \circ T_{2i}, & \text{if } i \leq j \\ T_{2i}, & \text{if } i > j. \end{cases}$$

Proof. We have the following pullback diagram

$$\begin{array}{ccc} \mathcal{C}|_{\Delta^{k-j}} & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow \\ \Delta^{k-j} & \longrightarrow & \Delta^k \end{array}$$

with the bottom map

$$\Delta^{k-j} \rightarrow \Delta^k \\ (t_1, \dots, t_j, t_{j+1}, \dots, t_{k-j}) \mapsto (t_1, t_1, \dots, t_j, t_j, t_{j+1}, t_{j+2}, \dots, t_k).$$

Then by Proposition 2.19, it follows that the monodromy operators of the universal deformation $\mathcal{C}|_{\Delta^{k-j}} \rightarrow \Delta^{k-j}$ are of the desired shape. $\square$

Lemma 5.6. *The monodromy weight filtration of the degeneration of Prym varieties above has the shape*

$$\begin{aligned} W_{-2}H_1(P_t, \mathbb{Z}) &= \text{Van}_{C_t}^- \\ W_{-1}H_1(P_t, \mathbb{Z}) &= (\text{Van}_{C_t}^-)^{\perp_{\Xi_t}} \\ W_0H_1(P_t, \mathbb{Z}) &\cong H_1(C_t, \mathbb{Z})^-, \end{aligned}$$

where by $(-)^-$ we mean the -1 eigenspace of the induced involution on the $\mathbb{Z}$ -module.

Proof. The isomorphism $W_0H_1(P_t, \mathbb{Z}) = H_1(P_t, \mathbb{Z}) \cong H_1(C_t, \mathbb{Z})^-$ follows by the characterization (3). Define $N^p := \sum_{i=1}^{k-j} N_i^p$ where for $i \leq j$, $N_i^p = N_{2i} + N_{2i-1}$ and for $i > j$, $N_i^p = N_{2i}$. Thus $N^p = N^- := N|_{H_1(C_t, \mathbb{Z})^-}$ giving us the isomorphism on the -2 piece. The isomorphism on the -1 piece follows from [Sch73, Lemma 6.4]. $\square$

Similar to the case of degenerations of Jacobians, we want to give a representation of $\mathrm{gr}_0^W H_1(P_t, \mathbb{Z})$ in terms of the dual graph of the central fiber of the degeneration. Consider the exact sequence

$$0 \longrightarrow W_{-1}H_1(C_t, \mathbb{Z}) \longrightarrow W_0H_1(C_t, \mathbb{Z}) \longrightarrow \mathrm{gr}_0^W H_1(C_t, \mathbb{Z}) \longrightarrow 0$$

Let $G = \mathbb{Z}/2\mathbb{Z}$ act on the modules above by the involution $\sigma = -\iota$. The sequence is an exact sequence of G -modules. Applying the basic theory of group cohomology, we obtain an exact sequence

$$0 \longrightarrow (\mathrm{Van}_{C_t}^{\perp \Theta_t})^- \longrightarrow H_1(C_t, \mathbb{Z})^- \longrightarrow (\mathrm{gr}_0^W H_1(C_t, \mathbb{Z}))^- \xrightarrow{\delta} H^1(G, \mathrm{Van}_{C_t}^{\perp \Theta_t})$$

We make a few observations:

- (1) By Lemma 3.4 and the identity $2\Xi = \Theta|_{P(f)}$, we have that

$$(\mathrm{Van}_{C_t}^{\perp \Theta_t})^- = (\mathrm{Van}_{C_t}^-)^{\perp \Xi_t}$$

- (2) By Proposition 4.16, we have an isomorphism

$$(\mathrm{gr}_0^W H_1(C_t, \mathbb{Z}))^- \cong H_1(\Gamma(C_0), \mathbb{Z})^-.$$

- (3) Finally, the group $H^1(G, \mathrm{Van}_{C_t}^{\perp \Theta_t})$ has the form

$$\frac{\ker(1 - \iota)}{\mathrm{im}(1 + \iota)},$$

where both maps are defined on the module $\mathrm{Van}_{C_t}^{\perp \Theta_t}$ (see [RR09, Thm. 9.27]).

It is clear this is a 2-torsion module.

We obtain a canonical isomorphism

$$\mathrm{gr}_0^W H_1(P_t, \mathbb{Z}) \cong \ker \delta.$$

We want to characterize the module $\ker \delta$ purely in terms of the combinatorial data obtained from the central fiber of the degeneration.

5.3. Characterization of $\ker \delta$. The aim of this subsection is to prove the following characterization of the module $\ker \delta$.

Theorem 5.7. *Suppose we have an admissible double cover of stable curves $C_0 \rightarrow C'_0$. Then a simple anti-invariant cycle in the module $H_1(\Gamma(C_0), \mathbb{Z})^-$ is an element of $\ker \delta$ if and only if it does not pass through a fixed vertex with no branchwise fixed node incident to it.*

To prove this theorem, we need some preparations.

Remark 5.8. From now on, we assume an orientation on the dual graph such that the involution always reverses the orientation, except for the branchwise fixed nodes for which this is impossible. What we mean by this is that if two edges e, e' of the dual graph are exchanged under the involution then we choose an orientation so that $e \mapsto -e'$.

Lemma 5.9. *Let G be a connected graph and T a spanning tree of G . Color each edge of $G \setminus T$ red and suppose we have the rule: “if v is a vertex of G with n edges incident to it such that $n - 1$ edges incident to v are red, then the final edge connected to v is also red”. With this rule each edge in the graph is red.*

Proof. Assume the spanning tree T has multiple vertices. Then T has at least two leaves. These can thus be colored red, then remove these two from the tree. Repeat until our spanning tree is empty and the whole graph is red. $\square$

Proposition 5.10. *Let C_0 be a stable curve and suppose choose a basis of $H_1(\Gamma(C_0), \mathbb{Z})$ obtained from a spanning tree $T \subseteq \Gamma(C_0)$. Then the vanishing cycles corresponding to the edges of $\Gamma(C_0) \setminus T$ form a basis of the module $\text{Van}_{C_t} \subseteq H_1(C_t, \mathbb{Z})$.*

Proof. Let $v \in \Gamma(C_0)$ be a vertex. Then the sum of all the vanishing cycles corresponding to edges incident to v is equal to a boundary (where the sum is compatibly chosen with the orientation of the vanishing cycles) and is thus trivial in $H_1(C_t, \mathbb{Z})$. We conclude by Lemma 5.9. For linear independence, we use

$$\begin{aligned} \text{rk}_{\mathbb{Z}}(\text{Van}_{C_t}) &= \text{rk}_{\mathbb{Z}}(H_1(C_t, \mathbb{Z})) - \text{rk}_{\mathbb{Z}}(H_1(C_0, \mathbb{Z})) \\ &= \text{rk}_{\mathbb{Z}}(H_1(\Gamma(C_0), \mathbb{Z})) \\ &= |\Gamma(C_0) \setminus T|. \end{aligned}$$

This finishes the proof. $\square$

Lemma 5.11. *Suppose $C_0 \rightarrow C'_0$ is an admissible double cover. If the induced involution on the dual graph acts freely on the vertices, then $\ker \delta = H_1(\Gamma(C_0), \mathbb{Z})^-$.*

Proof. Let γ be an anti-invariant cycle in the dual graph. We have the exact sequence

$$0 \longrightarrow H_1(C_0^\nu, \mathbb{Z}) \longrightarrow H_1(C_0, \mathbb{Z}) \longrightarrow H_1(\Gamma(C_0), \mathbb{Z}) \longrightarrow 0$$

following from the normalization sequence. Because ι acts freely on the vertices, it induces a pairing on the vertices of the dual graph, and thus we may decompose the homology of the normalization as

$$\bigoplus_j H_1(A_j, \mathbb{Z}) \oplus H_1(A_{\iota(j)}, \mathbb{Z})$$

where $A_j \hookrightarrow C_0^\nu$ is a connected component of the normalization. The involution sends a class $x \in H_1(A_j, \mathbb{Z})$ to $\iota(x) \in H_1(A_{\iota(j)}, \mathbb{Z})$ and we may thus decompose an invariant cycle as $y + \iota(y)$. Thus $H^1(G, H_1(C_0^\nu, \mathbb{Z})) = 0$ and consequently $H_1(C_0, \mathbb{Z})^- \rightarrow H_1(\Gamma(C_0), \mathbb{Z})^-$ is surjective. We will now use the exact sequence from Corollary 2.14

$$0 \longrightarrow \text{Van}_{C_t} \longrightarrow H_1(C_t, \mathbb{Z}) \longrightarrow H_1(C_0, \mathbb{Z}) \longrightarrow 0$$

where we used the isomorphism $H_1(C_0, \mathbb{Z}) \cong H_1(\mathcal{C}, \mathbb{Z})$ from Proposition 2.12. Let $T \subseteq \Gamma(C_0)$ be a spanning tree of the quotient. This gives two disjoint trees, T_1, T_2 in $\Gamma(C_0)$ such that $\iota(T_1) = T_2$. We can connect these two by a single edge, $\bar{e}$, and $T := T_1 \cup T_2 \cup \{\bar{e}\}$ is a spanning tree for $\Gamma(C_0)$. Then we get a basis for Van_{C_t} such

that ι acts on it as

$$\begin{pmatrix} 0 & -1 & & & v_1 \\ -1 & 0 & & & \vdots \\ & & \ddots & & \\ & & & 0 & -1 \\ & & & -1 & 0 & \vdots \\ 0 & \cdots & & 0 & v_n \end{pmatrix}.$$

If $v_n \neq 1$, we are done, as then there are no invariant combinations of vanishing cycles, meaning $H^1(G, \text{Van}_{C_t}) = 0$. If $v_n = 1$, then $H^1(G, \text{Van}_{C_t}) = \langle \beta \rangle / 2\langle \beta \rangle$ where

$$\beta = (\gamma_e \mid e \in \Gamma(C_0) \setminus T) \cdot \begin{pmatrix} v_1 & 0 & v_3 & 0 & \cdots & v_{n-1} & 0 & 1 \end{pmatrix}^\top.$$

We will argue in two cases. First, suppose $\alpha \in H_1(\Gamma(C_0), \mathbb{Z})$ such that $x_{\bar{e}}(\alpha) = 0$, then $\alpha = \alpha_1 + \alpha_2$ with $\alpha_i \in \Gamma_i(C_0)$ which are the subgraphs of $\Gamma(C_0)$ on the vertices of T_i . Moreover, $\iota(\alpha_1) = -\alpha_2$ meaning we can write $\alpha = \alpha_1 - \iota(\alpha_1)$. Then there is an $\alpha'_1 \in H_1(C_0, \mathbb{Z})$ such that $\alpha' := \alpha'_1 - \iota(\alpha'_1) \mapsto \alpha_1 - \iota(\alpha_1)$ and this is thus $\alpha' \in H_1(C_0, \mathbb{Z})$. Then similarly, α'_1 defines an element $\alpha''_1 \in H_1(C_t, \mathbb{Z})$ and $\alpha''_1 := \alpha'_1 - \iota_0(\alpha'_1)$ is an anti-invariant element in $H_1(C_t, \mathbb{Z})$ and a preimage of α . Thus $\delta(\alpha') = 0$.

Now suppose that $x_{\bar{e}}(\alpha) \neq 0$ and let α' be the lift in $H_1(C_0, \mathbb{Z})^-$. Then

$$\begin{aligned} 0 &= \Theta_t(\beta, \delta(\alpha')) = \left(\sum_e v_e \Theta_t(\gamma_e, \delta(\alpha)) \right) + \Theta_t(\gamma_{\bar{e}}, \delta(\alpha)) \\ &= \left(\sum_e v_e (\Theta_t(\gamma_e, \alpha') + \Theta_t(\gamma_e, \iota(\alpha'))) \right) + \Theta_t(\gamma_{\bar{e}}, \delta(\alpha)) \\ &= 2\Theta_t(\gamma_{\bar{e}}, \alpha') \neq 0 \end{aligned}$$

Here we used repeatedly that $\Theta_t(x, y) = \Theta_t(\iota(x), \iota(y))$ and that β is invariant. This means that $\delta(\alpha')$ and β intersect non-trivially meaning they cannot be equal, and thus the class $\delta(\alpha)$ is trivial in $H^1(G, \text{Van}_{C_t})$. We may therefore find a preimage of α' in the module $H_1(C_t, \mathbb{Z})^-$. $\square$

Lemma 5.12. *Let C be a smooth connected curve, ι a fixed point free involution on C and $x \in C$ a point. If γ is a path from x to $\iota(x)$, then the class $[\gamma \cdot \iota(\gamma)] \in H_1(C, \mathbb{Z})$ is nonzero.*⁵

Proof. First of all, notice that the paths γ and $\iota(\gamma)$ both define the same loop in the quotient C/ι , which we will denote by α . Moreover, notice that $[\gamma \cdot \iota(\gamma)] \in \pi_1(C, x)$ is invariant under ι and thus by the injection $H_1(C, \mathbb{Z})^{\{\iota=1\}} \hookrightarrow H_1(C/\iota, \mathbb{Z})$ (which follows from $H_k(X/G, \mathbb{Q}) \cong H_k(X, \mathbb{Q})^G$ for a smooth projective variety X and a finite group G acting freely on X , together with the fact that there is no torsion), it suffices to show that its image, which is $2[\alpha]$, is nonzero. This is equivalent to showing that $[\alpha]$ is nonzero. To show that $[\alpha]$ is nonzero, we can equivalently show that it does not define a commutator in the group $\pi_1(C/\iota, x)$. By the Galois correspondence, the

⁵Thank you Chris Vos and Marcel Masque for collaborating with me on this cute problem.

subgroup $f_*(\pi_1(C, x)) \subseteq \pi_1(C/\iota, x)$ is of index two, and thus the quotient is of order two which means it is abelian. Then if α were a commutator, its image in the quotient $\pi_1(C/\iota, x)/f_*(\pi_1(C, x))$ would be trivial, meaning it lies in $f_*(\pi_1(C, x))$. Then there would be a loop in C mapping to α , and by the homotopy lifting property, this loop would be homotopic to either γ or $\iota(\gamma)$. This is a contradiction since they are both just paths by freeness of the involution. $\square$

Lemma 5.13. *Let $C_0 \rightarrow C'_0$ be an admissible double cover of stable curves. Then any simple connected anti-invariant cycle γ passing through a fixed vertex is not an element of the kernel of the connecting map on the group cohomology $\delta : H_1(\Gamma(C_0), \mathbb{Z})^- \rightarrow H^1(G, \text{Van}_{C_t}^\perp)$.*

Proof. Suppose γ is such a cycle and consider the path it defines on the irreducible component A corresponding to the fixed vertex. Then $\iota(\gamma)$ defines also a path on A , and their concatenation defines an invariant cycle in the module $H_1(A, \mathbb{Z})$, say α . By Lemma 5.12, this cycle is non-trivial and thus the cycle α , when viewed in the module $H_1(C_t, \mathbb{Z})$, is also nonzero and does not intersect any vanishing cycle. Thus α is an invariant element of $\text{Van}_{C_t}^\perp$ and therefore defines a non-trivial class in $H^1(G, \text{Van}_{C_t}^\perp)$. It is clear that the map δ sends γ to α plus other terms linearly independent of α . Since α is nonzero, $\gamma \notin \ker \delta$. $\square$

Along similar lines, one may show that if a branchwise fixed node is attached to a fixed node, a simple ant-invariant cycle passing through it is an element of $\ker \delta$. We will only give sketches of some of the proofs.

Lemma 5.14. *Let C be a smooth connected curve, ι an involution with $e \in C$ a point fixed by ι . If $x \in C$ is any nonfixed point of C , then we may find a path γ from x to $\iota(x)$ such that it is homotopic to $\overline{\iota(\gamma)}$.*

Proof. In C/ι take a simply connected open neighbourhood U containing x and e and pick a path α from x to e in U . The preimage of α along $f : C \rightarrow C/\iota$ defines a path γ from x to $\iota(x)$ and contains the fixed point e . Clearly $\iota(\gamma) = \bar{\gamma}$ and thus $\gamma = \overline{\iota(\gamma)}$. $\square$

Lemma 5.15. *Suppose we have an admissible double cover of stable curves $C_0 \rightarrow C'_0$. Let $\gamma \in H_1(\Gamma(C_0), \mathbb{Z})^-$ be a basis element such that if it passes through a vertex that is fixed, then this vertex must have a branchwise fixed node incident to it. Then $\gamma \in \ker \delta$.*

Proof (sketch). Using Lemma 5.14, one sees $H_1(C_0, \mathbb{Z})^- \rightarrow H_1(\Gamma(C_0), \mathbb{Z})^-$ is surjective. To show that $H_1(C_t, \mathbb{Z})^- \rightarrow H_1(C_0, \mathbb{Z})^-$ is surjective, notice that the only obstruction can be obtained from the vanishing cycles attached to the branchwise fixed nodes. One uses the Riemann-Hurwitz formula to show that the branchwise fixed nodes form a closed cycle in $H_1(\Gamma(C_0), \mathbb{Z})$ and by Lemma 5.9, the vanishing cycles attached to fixed nodes will be linearly dependent in Van_{C_t} , which will prove the lemma. $\square$

Proof (Theorem 5.7). By Lemma 5.13 and 5.15, it suffices to show the reverse direction. Let $(C_0)_{\text{free}} \hookrightarrow C_0$ be the subcurve where we remove the fixed irreducible components. Then given such an anti-invariant cycle, it defines an element of $H_1(\Gamma((C_0)_{\text{free}}, \mathbb{Z})^-$

which by Lemma 5.11, we can pull back to an element of $H_1((C_t)_{\text{free}}, \mathbb{Z})^-$ which we can push forward⁶ to an element of $H_1(C_t, \mathbb{Z})^-$ proving the theorem. $\square$

5.4. Monodromy bilinear forms degenerations of Prym varieties. Recall that we are analyzing the degeneration of Prym varieties $P^* \rightarrow (\Delta^*)^{k-j}$ coming from an admissible double cover of stable curves $C_0 \rightarrow C'_0$ restricted to the slice where the monodromy is non-trivial.

Lemma 5.16. *The monodromy bilinear forms on $\text{gr}_0^W H_1(P_t, \mathbb{Z})$ have the shape*

$$B_e(x, y) = \begin{cases} \Theta_t(x, \gamma_e) \Theta_t(y, \gamma_e), & \text{if } e \text{ is non-fixed,} \\ 0 & \text{if } e \text{ is branchwise fixed.} \end{cases}$$

where by e we mean the hyperplane it defines in the degeneration $P^* \rightarrow (\Delta^*)^{k-j}$.

Proof. First of all, for e non-fixed, we have

$$\begin{aligned} B_e(x, y) &= \Xi_t(N'_e(x), y) \\ &= \frac{1}{2} \Theta_t(N_e(x) + N_{\iota(e)}(x), y) \\ &= \frac{1}{2} (\Theta_t(x, \gamma_e) \Theta_t(y, \gamma_e) + \Theta_t(x, \gamma_{\iota(e)}) \Theta_t(y, \gamma_{\iota(e)})), \\ &= \frac{1}{2} (\Theta_t(x, \gamma_e) \Theta_t(y, \gamma_e) + \Theta_t(\iota(x), \iota(\gamma_{\iota(e)})) \Theta_t(\iota(y), \iota(\gamma_{\iota(e)}))), \\ &= \Theta_t(x, \gamma_e) \Theta_t(y, \gamma_e). \end{aligned}$$

Now let e be a branchwise fixed node. Then $\Theta_t(x, y) = \Theta_t(\iota(x), \iota(y))$ and notice that for any anti-invariant cycle x , we get $\Theta_t(x, \gamma_e) = \Theta(\iota(x), \gamma_e) = -\Theta_t(x, \gamma_e)$. This can only be true if $\Theta_t(-, \gamma_e) = 0$ on $H_1(C_t, \mathbb{Z})^-$. $\square$

We obtain the following:

Theorem 5.17. *Let $C_0 \rightarrow C'_0$ be an admissible double cover of curves and consider the corresponding degeneration of Prym varieties $P^* \rightarrow (\Delta^*)^{k-j}$. The monodromy bilinear forms are of the shape $B_e = x_e^2$ where x_e is the coordinate projection on $\ker \delta$ restricted along the chain of inclusions*

$$\ker \delta \subseteq H_1(\Gamma(C_0), \mathbb{Z})^- \subseteq H_1(\Gamma(C_0), \mathbb{Z}) \subseteq \mathbb{Z}^{E(\Gamma(C_0))}.$$

Moreover, if the map

$$\begin{aligned} E(\Gamma(C_0))/\iota &\rightarrow \ker \delta^\vee \\ [e] &\mapsto x_e \end{aligned}$$

forms an integral realization of a regular matroid, the monodromy cone is matroidal.

Proof. Because

$$\text{gr}_0^W H_1(P_t, \mathbb{Z}) \cong \ker \delta \subseteq H_1(\Gamma(C_0), \mathbb{Z}) \cong \text{gr}_0^W H_1(\mathcal{J}C_t, \mathbb{Z}),$$

⁶Note that a map $(C_t)_{\text{free}} \rightarrow C_t$ may not exist, but the map $H_1((C_t)_{\text{free}}, \mathbb{Z}) \rightarrow H_1(C_t, \mathbb{Z})$ does by surjectivity of $H_1(C_t, \mathbb{Z}) \rightarrow H_1(C_0, \mathbb{Z})$ and the map $H_1((C_0)_{\text{free}}, \mathbb{Z}) \rightarrow H_1(C_0, \mathbb{Z})$.

the forms $x \mapsto \Theta_t(x, \gamma_e)$ again correspond to the coordinate projections $x_e : \mathbb{Z}^{E(\Gamma(C_0))} \rightarrow \mathbb{Z}$ but now restricted to $\ker \delta$ by Proposition 4.16 and Lemma 5.16.

Indeed, if e is a branchwise fixed node, any anti-invariant cycle in the dual graph cannot pass through e because the orientation must be preserved under the involution, meaning x_e is zero on $H_1(\Gamma(C_0), \mathbb{Z})^-$. Notice moreover that, with the orientation as in Remark 5.8, the forms $x_e = x_{e'}$ on $H_1(\Gamma(C_0), \mathbb{Z})^-$ so the map

$$E(\Gamma(C_0)) \rightarrow \ker \delta^\vee,$$

is well defined. □

6. SIGNED GRAPHS

We will start with a short treatment of all the necessities of signed graphs which are needed later. For an incredibly good introduction to signed graphs we refer to [Zas82]. Almost everything which we will state here can be found in the aforementioned article.

Definition 6.1 ([Zas82]). *A signed graph is a pair $\Sigma = (G, \sigma)$ consisting of a graph G together with a sign function $\sigma : E(G) \rightarrow \{\pm 1\}$. By the notation $e : vw$ we mean an edge $e \in E(G)$ between vertices v and w in $V(G)$. A loop will mean an edge $e : vv$ and a link $e : vw$ with v and w distinct.*

Remark 6.2. Signed graphs may have half edges (edges incident to a single vertex) and free loops (edges incident to no vertices). To study general double covers of graphs, one needs this formalism, but all double covers of graphs we will encounter do not need these notions as we will consider admissible double covers and, at a later stage, double covers with no fixed vertices.

Definition 6.3 ([Zas82]). *Let Σ be a signed graph. We say a cycle γ is negative if $\prod_{e \in \gamma} \sigma(e) = -1$. A signed graph is balanced if it does not contain any negative cycles. We say a vertex $v \in V(G)$ is a balancing vertex if removing it from the graph will result in a balanced signed graph.*

On a signed graph we may perform an operation called *switching*, which will be invariant on the upcoming matroids.

Definition 6.4 ([Zas82]). *Let Σ be a signed graph and take a vertex $v \in V(G)$ of the underlying graph. Switching at v means that we multiply all the signs of the links incident to v by -1 .*

Minors of signed graphs are defined similar as for graphs.

Definition 6.5 ([Zas82]). *Deletion and contraction of a signed graph are defined the same as for ordinary graphs, except that contraction of negative links is first defined via switching at either vertex of said link. A signed graph Σ' is a minor of another signed graph Σ if it can be obtained from the operations switching, contraction and deletion.*

To a signed graph we may associate two matroids: the signed-graphic matroid and, its dual, the cosigned-graphic matroid⁷.

Definition 6.6 ([Zas82]). *Let Σ be a signed graph. On the edge set of the underlying graph, $E(G)$, denote by $M(\Sigma)$ the signed-graphic matroid defined as follows: a subset of $E(G)$ is independent if and only if each of its connected components contains at most one cycle and that one negative. Any matroid which is dual to a signed-graphic matroid is called a cosigned-graphic matroid.*

Theorem 6.7 ([Zas82, Cor. 5.4]). *Let G be a graph and σ, σ' be two sign functions. Denote by Σ and Σ' the corresponding signed graphs. Then $M(\Sigma) = M(\Sigma')$ if and only if Σ' can be obtained from Σ by switching.*

Signed-graphic matroids form a generalization of graphic matroids in the following sense.

Theorem 6.8 ([Zas82, Prop. 7A.1]). *Let G be a graph and $+$ be the sign function that always assigns $+$ to an edge. Denote by $+G := (G, +)$. Then $M(+G) = M(G)$ where the left is the signed-graphic matroid and the right is the standard graphic matroid.*

We will now discuss representability of signed-graphic matroids.

Definition 6.9 ([Zas82]). *Let Σ be a signed graph. We define the incidence matrix $M_\Sigma : \mathbb{Z}^{E(G)} \rightarrow \mathbb{Z}^{V(G)}$ to be the linear map $e \mapsto M_e = (m_{ve} \mid v \in V(G))$ defined by the rules*

$$\begin{aligned} m_{ev} &= \pm 1 \text{ and } m_{ew} = -m_{ev}\sigma(e), \text{ if } e : vw \text{ with } v, w \text{ distinct} \\ m_{ev} &= 0, \text{ if } e : vv \text{ a loop and } \sigma(e) = +1 \\ m_{ev} &= \pm 2, \text{ if } e : vv \text{ a loop and } \sigma(e) = -1 \end{aligned}$$

The columns M_e of the incidence matrix come in pairs: one the negative of the other. Which one you choose corresponds to an orientation of the edges. This is why we speak of *the* incidence matrix, since changing the sign of an entire column is immaterial to us.

Theorem 6.10 ([Zas82, Prop. 8A.5]). *Let Σ be a signed graph. Then the incidence matrix M_Σ is totally unimodular if and only if Σ is balanced.*

Theorem 6.11 ([Zas82, Thm. 8B.1]). *Let Σ be a signed graph and F any field of characteristic not equal to two. Then the incidence matrix M_Σ represents $M(\Sigma)$ over F .*

The following theorem can be derived from [AK06, Thm. 24]:

Theorem 6.12 ([AK06, Thm. 24]). *Let Σ be a signed graph and $\widetilde{M}_\Sigma$ a full row rank submatrix of the incidence matrix M_Σ . Let R be an invertible square submatrix of $\widetilde{M}_\Sigma$*

⁷The name signed-cographic matroid seems more appropriate, but we will stick to the terminology of [P⁺14].

also of full row rank and write $\widetilde{M}_\Sigma = [R \mid S]$. Then $[I \mid R^{-1}S]$ is a totally unimodular matrix representation of $M(\Sigma)$ if Σ is balanced or if every basis of Σ is connected.

We will now see which signed-graphic matroids are regular.

Definition 6.13. Let Σ be a signed graph. We will call Σ *tangled* if it is not balanced nor does it contain a balancing vertex and there cannot be two vertex-disjoint negative cycles.

Observe that if Σ is connected and tangled, then every basis is connected by definition.

Theorem 6.14. [SQ07] Let Σ be a signed graph. Then $M(\Sigma)$ is regular if and only if

- (1) $M(\Sigma)$ is a graphic matroid or
- (2) Σ is tangled.

Remark 6.15. Notice that if $M(\Sigma)$ is a regular matroid, there is no guarantee that $[I \mid R^{-1}S]$ is totally unimodular. See Example 7.11.

The following proposition is well-known in the theory of signed graphs.

Proposition 6.16. Let $-K_n$ be the complete signed graph on n vertices with all edges negative. We have the following:

- (1) $M^*(-K_n)$ for $n = 1, 2, 3, 4$ are cographic;
- (2) $M^*(-K_5) \cong R_{10}$;
- (3) $M^*(-K_n)$ for $n > 5$ are not regular.

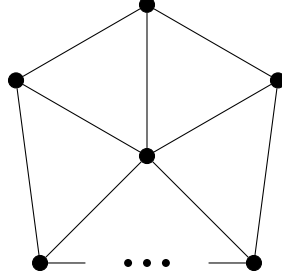
Proof. For $n = 1$ there is nothing to prove. For $n = 2$ we can switch at a vertex and it is immediate that $M(-K_2) \cong M(K_2)$. For $n = 3$ we observe that it has a balancing vertex: removing any vertex will result in $-K_2$ which is balanced. This means that $-K_3$ is graphic [Sli05, pp. 9]. For $n = 4$ one sees that $-K_4$ has precisely 3 circuits of length 4 from which it is easy to construct an explicit isomorphism to $K_{2,3}$ to conclude $M(-K_4) \cong M(K_{2,3})$. The result for $n = 5$ can be found in the literature, for example [P⁺14] or [QSZ09].

Any signed graph $-K_n$ with $n > 5$ has two disjoint negative cycles meaning it is not tangled. It can also not be graphic, because it contains R_{10} as a minor. $\square$

Example 6.17. Let $-W_5$ be the wheel graph with a vertex in the center and a 5 vertex rim around the outside. It is shown in [Zas90, Proposition 4A] that $M(-W_5) \cong M^*(K_5)$ and consequently $M^*(-W_5) \cong M(K_5)$.

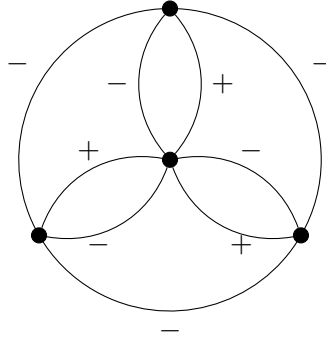
Example 6.18. Let $-W_n$ be a wheel graph. Notice that these are either graphic or tangled, meaning that $M^*(-W_n)$ is regular. It turns out that $-W_n$ is a minor of $-W_{n+2}$ which we will proceed to show now. Starting in $-W_{n+2}$, switch at a vertex in the rim, making it incident to only positive edges and contract the two side edges contained in the rim. What we are left with is $-W_n$ with 3 edges between the center vertex and the vertex we switched at, two are negative and one is positive. Simply delete the

positive and one negative edge to arrive at $-W_n$. This means that $M(-W_n)$ is a minor of $M(-W_{n+2})$. Equivalently, $M^*(-W_n)$ is a minor of $M^*(-W_{n+2})$. In particular, notice that for all $n \geq 7$ odd, $M(K_5)$ is a minor of $M^*(-W_n)$ which means that the negative, odd, cosigned-graphic wheel matroids are not cographic. A visualization of wheel graphs is depicted below.



Example 6.19. One might wonder what happens for even wheel graphs. Let $-W_n$ be the wheel graph with n even. Then the center is a balancing vertex. By [Sli05, pp. 9] we know $M(-W_n)$ is graphic, and $M^*(-W_n)$ is therefore cographic.

Example 6.20. The graphic matroid of the complete bipartite on 6 vertices $M(K_{3,3})$ can be realized as a cosigned-graphic matroid (see [P⁺14, Fig. 6.9]. Its signed graph is depicted below:



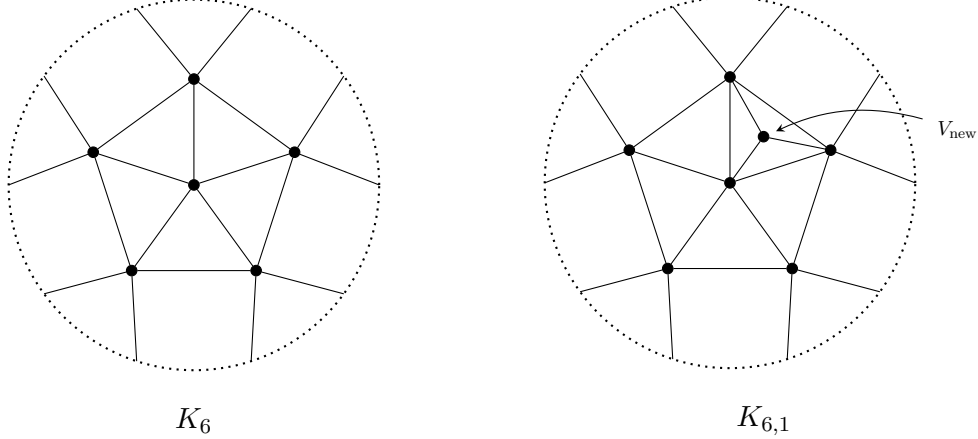
Example 6.21. The following example is an application of [Sli05] and will be used for the proof of Theorem 1.4. The main idea is that if a graph G can be embedded in the projective plane $\mathbb{RP}^2$, which is called a *projective-planar graph*, then one may associate a sign function σ on its *projective-planar dual graph* G^* such that $M^*(G) \cong M^*(G^*, \sigma)$.

Let G be a projective-planar graph. The projective-planar dual graph G^* of G can be constructed as follows: replace each face of the graph by a vertex and for every edge in the original graph, draw an edge between the vertices corresponding to the faces on both sides of that edge. The sign function σ on G^* is constructed as follows: let γ be a cycle in G that is *non-separating*, meaning that the curve it defines in the projective plane has the property that $\mathbb{RP}^2 \setminus \gamma$ is connected. Then all edges in G^* corresponding to an edge in the cycle C will be signed negative and the remaining will be positive. The sign function is defined up to switching (see [Sli05, pp. 2] for more details).

Let K_6 be the complete graph on 6 vertices, which is not a planar graph, but is projective-planar (see the figure below). Consider its projective-planar dual K_6^* with its sign function σ . By [Sli05, Thm. 2] we have an isomorphism $M^*(K_6) \cong M(K_6^*, \sigma)$

and by [Sli05, Thm. 7], the signed graph (K_6^*, σ) is tangled.

We can generalize this example inductively. Notice that all faces of K_6 are triangles. Let $K_{6,0} := K_6$ and define $K_{6,n+1}$ to be the graph $K_{6,n}$ and adding a vertex V_{new} in a face F of $K_{6,n}$ and draw a single edge between the new vertex V_{new} and to each vertex in the face F . See the illustration for $K_{6,1}$ from K_6 below.



In particular notice that, by simply deleting the new vertex added, $K_{6,n}$ is a minor of $K_{6,n+1}$, so that they are all non-planar but projective-planar. We moreover have that $|V(K_{6,n})| = 6 + n$ and $|E(K_{6,n})| = 15 + 3n$, so that

$$\text{rk}(M(K_{6,n})) = 5 + n.$$

In particular, by applying [Sli05, Thm. 2] and [Sli05, Thm. 7] again, we obtain for every $n \geq 0$, a tangled signed graph Σ on $15 + 3n$ edges such that

$$M^*(\Sigma) \cong M(K_{6,n})$$

which is of rank $5 + n$ and not cographic, as it contains $M(K_6)$ as a minor.

7. SIGNED GRAPHS AND PRYM VARIETIES

With the theory of signed graphs at our disposal, we can connect this to our previous discussion on Prym varieties. Namely, given a graph G with an involution ι on it, we want to analyze if the matrix

$$\varphi : E(G)/\iota \rightarrow (H_1(G, \mathbb{Z})^-)^\vee$$

realizes a regular matroid. The idea now is that we construct a signed graph such that its cosigned-graphic matroid remembers the double cover structure and is represented by φ . This idea was originally due to Zaslavsky [Zas82], and the connection to tropical Prym varieties was established in [RZ23]. That φ realizes cosigned-graphic matroids, however, appears to be new.

Let G be a connected graph with an admissible involution ι that acts freely on the vertices. Denote by G/ι the quotient. Partition $V(G)$ into two disjoint sets V^+ and V^- where $\iota(V^\pm) = V^\mp$. We call $V^\pm$ the *sheets* of G and we say an edge in G *exchanges sheets* if it is incident to both a vertex in V^+ and in V^- . Notice that e exchanges sheets if and only if $\iota(e)$ does. We define a sign function on G/ι :

*An edge has sign -1 if and only if its preimages exchange sheets.*⁸

Remark 7.1. The sign function is only defined up to switching. Indeed for each $v \in V(G/\iota)$, swapping the vertices in the sheets results in switching at the vertex $v \in V(G/\iota)$. This will be fine for our purposes by Theorem 6.7.

Denote the resulting signed graph by Σ_ι on the graph G/ι . We are now ready to state the main theorem of the master's thesis.

Theorem 7.2. *Let G be graph with an admissible involution ι acting freely on the vertices. Then the linear map*

$$\begin{aligned} \varphi : E(G)/\iota &\rightarrow (H_1(G, \mathbb{Z})^-)^\vee \\ [e] &\mapsto x_e \end{aligned}$$

by restricting coordinate projections is a near-integral realization of $M^(\Sigma_\iota)$. Moreover, this is an integral realization of $M^*(\Sigma_\iota)$ if Σ_ι is balanced or tangled.*

Proof. First of all, note that the obvious maps $C_i(G)^- \rightarrow C_i(G/\iota)$ are isomorphisms. We may form the following square:

$$(7) \quad \begin{array}{ccc} C_1(G)^- & \xrightarrow{\partial^-} & C_0(G)^- \\ \sim \downarrow & & \downarrow \sim \\ C_1(G/\iota) & \xrightarrow{M_{\Sigma_\iota}} & C_0(G/\iota) \end{array}$$

Choose a partition $V^\pm$ of $V(G)$ into sheets. The commutativity of diagram (7) follows from a diagram chase for each of the following instances:

- (1) pairs of links in G exchanging sheets and becoming a negative link in Σ_ι ;
- (2) pairs of links in G exchanging sheets and forming a negative loop in Σ_ι ;
- (3) pairs of links in G not exchanging sheets and forming a positive link in Σ_ι ;
- (4) pairs of loops in G forming a positive loop in Σ_ι .

Thus we get an isomorphism

$$\ker(M_{\Sigma_\iota}) \cong \ker(\partial^-).$$

We can compute $H_1(G, \mathbb{Z})^-$ as the kernel of the boundary operator acting on the -1 -eigenspaces of the chain groups

$$H_1(G, \mathbb{Z})^- = \ker(\partial^- : C_1(G)^- \rightarrow C_0(G)^-)$$

by commutativity of the boundary operator with the involution and because $(-)^{\mathbb{Z}/2\mathbb{Z}}$ is a left exact functor. Notice that the j 'th row of the matrix φ is the basis vector γ_j of $H_1(G, \mathbb{Z})^-$ and thus the rows of φ span the kernel of the incidence matrix by construction. It is a standard theorem that the dual matroid of the column matroid of a matrix is the column matroid of a matrix whose rows span the kernel of said matrix

⁸The original version that we defined only took the constructions of unsigned derived graphs into account, resulting in all negative signed graphs. By the article [RZ23], we were able to generalize this to the current version.

(see [Fes25, pp. 6]). What this means is that over all fields of characteristic not two:

$$\begin{aligned} M^*(\Sigma_\iota) &\cong \text{col}(M_{\Sigma_\iota})^* \\ &\cong \text{col}(\varphi). \end{aligned}$$

In other words, φ is a near-integral realization of the cosigned-graphic matroid $M^*(\Sigma_\iota)$. We now focus our attention on the regularity statement. From the matrix $\widetilde{M}_{\Sigma_\iota} = [R \mid S]$ we obtain $[I \mid R^{-1}S]$ which is a totally unimodular representation of $M(\Sigma_\iota)$ if Σ_ι is balanced or tangled by Theorem 6.12. Then

$$\begin{aligned} H_1(G, \mathbb{Z})^- &\cong \ker(\partial^-) \\ &\cong \ker(M_{\Sigma_\iota}) \\ &\cong \ker(\widetilde{M}_{\Sigma_\iota}) \\ &\cong \ker([I \mid R^{-1}S]). \end{aligned}$$

This means that $\varphi = U \cdot [-(R^{-1}S)^T \mid I]$ for some unimodular square matrix U . Thus φ represents $M^*(\Sigma_\iota)$ over all fields meaning it is an integral realization. $\square$

Remark 7.3. Theorem 7.2 can easily be generalized to any double cover of graphs using the notion of dilation as introduced in [RZ23] and using free loops and half edges. This would, however, only complicate our lives. Fixed irreducible components of an admissible double cover of curves would mean that $\ker \delta \neq H_1(\Gamma(C_0), \mathbb{Z})^-$.

Applying Theorem 5.17 together with Theorem 7.2, we obtain:

Theorem 7.4. *Let C_0 be a stable curve with an admissible involution ι_0 fixing no vertices of the dual graph. Then the monodromy cone of the degeneration of Prym varieties $P^* \rightarrow (\Delta^*)^{k-j}$ associated to it is near-matroidal with near-regular matroid $M^*(\Sigma_\iota)$. The monodromy cone is matroidal with matroid $M^*(\Sigma_\iota)$ if Σ_ι is balanced or tangled.*

Example 7.5. In Section 8, Gwena's stable curve with involution is constructed and we computed the matrix $\varphi : E(\Gamma(C_0))/\iota \rightarrow (H_1(\Gamma(C_0), \mathbb{Z})^-)^\vee$ giving us a representation of R_{10} . However, the result is much easier using Corollary 7.4: the signed graph Σ_ι is exactly $-K_5$, which is tangled, and the degeneration is maximal since all irreducible components have genus 0. We can thus conclude that the degeneration of Prym varieties $P^* \rightarrow (\Delta^*)^{10}$ is matroidal with matroid R_{10} .

7.1. Derived graphs. Let us now consider the reverse question. Namely, given a signed graph Σ , can we find an admissible double cover $C \rightarrow C'$ such that it is near-matroidal with matroid $M^*(\Sigma)$?

Definition 7.6. *Let $\Sigma = (G, \sigma)$ be a signed graph. We define the derived graph G_{dr} to be the graph with vertex set $V(G_{\text{dr}}) = V(G) \times \{0, 1\}$. The edge set $E(G_{\text{dr}})$ consists of two edges for every edge $e : vw$ in Σ : if it positive, the two edges are $e_i : v_i w_i$ for $i = 0, 1$ and if it is negative $e_i : v_i w_{1-i}$ for $i = 0, 1$.*

Remark 7.7. It is clear that G_{dr} is connected if and only if Σ is connected and unbalanced.

Remark 7.8. The definition of the derived graph takes origin in the theory of *voltage graphs* where one generalizes the sign function, in which case it will be called the *voltage*, to take values in any arbitrary group.

Theorem 7.9. *Given any connected unbalanced signed graph Σ , there is a degeneration of Prym varieties $P^* \rightarrow (\Delta^*)^{|E(\Sigma)|}$ such that its monodromy cone is near-matroidal with near-regular matroid $M^*(\Sigma)$ and matroidal if Σ is tangled.*

Proof. The derived graph construction for a signed graph will give a canonical double cover $G_{\text{dr}} \rightarrow G$ giving rise to a vertex- and edge-free involution on G_{dr} . To this double cover of graphs, we can associate an admissible double cover $C_0 \rightarrow C'_0$ of stable curves in the following way:

- (1) for each vertex $v_0 \in V(G_{\text{dr}})$ having exactly 1 edge incident, take a genus 2 curve and mark a point. For the vertex v_1 do the same;
- (2) for each vertex $v_0 \in V(G_{\text{dr}})$ having exactly 2 edges incident, take an elliptic curve and mark two distinct points. For the vertex v_1 paired with v_0 do the same;
- (3) for each vertex $v_0 \in V(G_{\text{dr}})$ having $a \geq 3$ edges incident, take a copy of $\mathbb{P}^1$ and mark a distinct points. For the vertex v_1 paired with v_0 do the same;
- (4) identify the marked points according to the edges;
- (5) define the involution by swapping the pairs of irreducible components corresponding to v_0 and v_1 .

Notice, in particular, that the double cover of the dual graphs of the curves is exactly $G_{\text{dr}} \rightarrow G$, meaning that $\Sigma_\iota = \Sigma$. Moreover the resulting nodes of C_0 are non-fixed and we have no smooth fixed points meaning it is an admissible double cover with $\ker(\delta) = H_1(\Gamma(C_0), \mathbb{Z})^-$. We thus obtain a degeneration of Prym varieties $P^* \rightarrow (\Delta^*)^{|E(\Sigma)|}$ with monodromy cone near-matroidal with matroid $M^*(\Sigma)$ and matroidal if Σ is tangled by Corollary 7.4. $\square$

We now have a large class of signed graphs to which we can associate a matroidal degeneration of Prym varieties, but perhaps not all. One asks the following question:

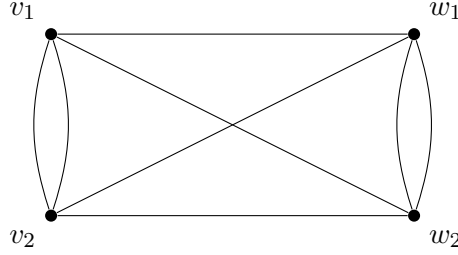
Question 1. *Given any signed graph Σ , when can we find a connected tangled signed graph Σ' such that their signed graphic matroids are isomorphic?*

A more general version of this question goes back all the way to Zaslavsky's original paper on signed graphs [Zas82, Problem 5.1]. It seems unanswered and hard.

7.2. Friedman-Smith Examples. We will now turn our attention to the Friedman-Smith examples. A stable curve C_0 with an admissible involution is a *Friedman-Smith example with $2n$ edges* if $C_0 = C_1 \cup C_2$ is the union of two components, both invariant under the involution, intersecting in $2n$ nodes, so that the involution exchanges these $2n$ nodes (see [Vol01] where they show that an admissible double cover in $\overline{R}_g$ is in the

indeterminacy locus of the map $\overline{R}_g \rightarrow \overline{\mathcal{A}}_{g-1}^V$ to the second Voronoi compactification if and only if it is a Friedman-Smith example with $2n \geq 4$ edges).

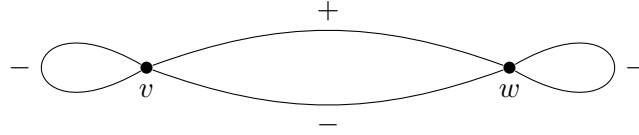
Example 7.10. Let $C_0 \rightarrow C'_0$ be an admissible double cover such that the dual graph $\Gamma(C_0)$ takes the shape:



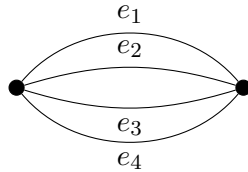
and the induced involution on $\Gamma(C_0)$ act as $v_1 \mapsto v_2$ and $w_1 \mapsto w_2$ and freely on the edges. This is a Friedman-Smith example with $2n = 4$ edges. Calculating the matrix $\varphi : \mathbb{Z}^{E(\Gamma(C_0/\iota))} \rightarrow (H_1(\Gamma(C_0), \mathbb{Z})^-)^\vee$ yields

$$(8) \quad \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{pmatrix}$$

which is a representation of $U_{2,4}$ over $\mathbb{R}$. It is easier, and allows us to conclude much faster that this cannot be a matroidal degeneration, to use Corollary 7.4. The signed graph Σ_ι has the shape



and it is standard signed graph theory that $M^*(\Sigma_\iota) = U_{2,4}^* = U_{2,4}$ (see for example [Zas97, Lemma 2.4]). The degeneration is therefore near-matroidal and not matroidal as $U_{2,4}$ is not regular. Notice moreover that $\Gamma(C_0)$ is precisely the derived graph of Σ_ι . An example very much related to the previous one, and one that shows why freeness of the vertices is important for Corollary 7.4, is the following.⁹ Let C_0 be the stable curve obtained by taking two copies of an elliptic curve E with a two torsion point $Q \in E[2]$ and identifying two pairs points that are Q away from each other. Let the involution act on the irreducible components as $P \mapsto P + Q$ to obtain a dual graph with an involution that fixes two vertices and swaps the pairs of edges e_1, e_2 and e_3, e_4 .



⁹We refer to [FS86] and [ABH01] and [CMGHL15] who all treat the same example in different ways, where we provide yet another direction.

The double cover $C_0 \rightarrow C'_0$ is clearly admissible, since we only have non-fixed nodes and no smooth fixed points and is clearly a Friedman-Smith example with $n = 2$. Orient the graph so that the orientation is reversed upon applying the involution. Take the spanning tree $\{e_1\}$ so that a basis of the homology is $H_1(\Gamma(C_0), \mathbb{Z}) = \langle e_1 + e_2, e_1 - e_3, e_1 + e_4 \rangle =: \langle \alpha_1, \alpha_2, \alpha_3 \rangle$. The involution acts as

$$\begin{aligned}\alpha_1 &\mapsto -\alpha_1 \\ \alpha_2 &\mapsto \alpha_3 - \alpha_1 \\ \alpha_3 &\mapsto \alpha_2 - \alpha_1\end{aligned}$$

meaning we have the basis for $H_1(\Gamma(C_0), \mathbb{Z})^- = \langle \alpha_1, \alpha_2 - \alpha_3 \rangle = \langle e_1 + e_2, e_3 + e_4 \rangle$. We wish to compute $\ker(\delta)$, and thus we need to compute $\delta(e_1 + e_2)$ and $\delta(e_3 + e_4)$. Let α'_1, α'_2 be the lifts of $e_1 + e_2$ and $e_3 + e_4$ respectively in $H_1(C_t, \mathbb{Z})$. It is clear that $\alpha'_1 + \iota(\alpha'_1) = \beta_2 \pmod{2}$ for both $i = 1, 2$. Therefore,

$$\ker(\delta) = \langle e_1 + e_2 + e_3 + e_4, e_1 + e_2 - e_3 - e_4 \rangle$$

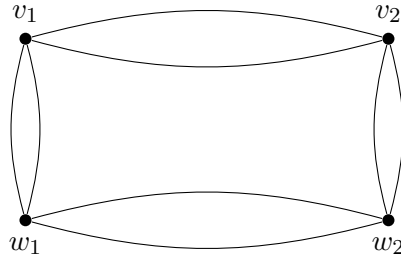
and thus the map $\mathbb{Z}^{E(\Gamma(C_0)/\iota)} \rightarrow \ker(\delta)^\vee$ is of the shape

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

which does not realize a regular matroid as the linear dependencies of the columns change when viewed over the fields $\mathbb{F}_2$ and $\mathbb{F}_3$. The monodromy cone is thus not matroidal.

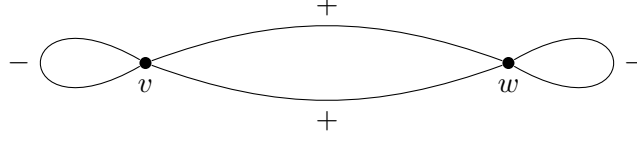
It is important to note that if we would have degenerated this example further, by adding a pair of extra vanishing cycles on each of the elliptic curves E lying Q away from each other, we would have arrived at the vertex-free example above. We cannot apply the theory of signed graphs to the latter example, because the vertices hide the fact that one half of the edges exchange sheets, whereas the other half does not.

Example 7.11. Let $C_0 \rightarrow C'_0$ be an admissible double cover of stable curves such that the dual graph $\Gamma(C_0)$ takes the shape:



and the involution swaps v_1 with v_2 and w_1 with w_2 and all parallel edges are swapped. This is another Friedman-Smith example with $n = 2$. The signed graph Σ_t looks as

follows:



Then it has exactly 3 circuits: two of length three and one of length two. Therefore $M(\Sigma) \cong M(G)$ where G is the complete graph on three vertices with an additional link between any two vertices. The matroid $M(\Sigma_\iota)$ is thus graphic, and therefore regular, and one would thus like to apply Theorem 7.2. This is not possible, however, as Σ_ι is not balanced and not tangled (the two negative loops). The issue comes from the fact that the matrix $[I \mid R^{-1}S]$ is not totally unimodular as one computes

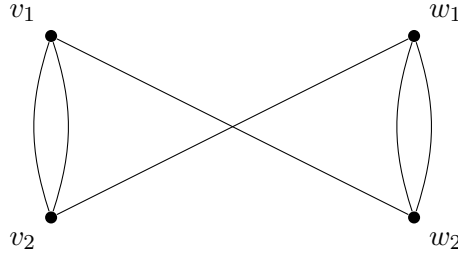
$$R^{-1}S = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} \end{pmatrix}.$$

Indeed, the monodromy cone is not matroidal as

$$\varphi = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 0 & 0 \end{pmatrix}$$

which contains a block with determinant -2 .

Example 7.12. Let us look at a Friedman-Smith example with $n = 1$. Take an admissible double cover of stable curves such that its dual graph has the shape



and the involution swaps v_1 with v_2 and w_1 with w_2 . By a similar argument as in Example 7.11, the matroid $M(\Sigma_\iota)$ is graphic and therefore regular, but Σ_ι is unbalanced and untangled. One expects thus that the monodromy cone will again not be matroidal, however computing φ gives the matrix

$$\varphi = \begin{pmatrix} 1 & 1 & 2 \end{pmatrix}.$$

Recall that the columns of φ represent the linear forms $x_{[e]}$ and thus the monodromy bilinear forms are

$$B_{[e_1]} = x_{[e_1]}^2, \quad B_{[e_2]} = x_{[e_2]}^2, \quad B_{[e_3]} = 4x_{[e_3]}^2$$

with $d_{[e_3]} = 4$. The forms $x_{[e_i]}$ thus realize a regular matroid. The monodromy cone of this degeneration is thus matroidal.

7.3. Applications to the integral Hodge conjecture for abelian varieties. We now proceed apply the theory developed to the integral Hodge conjecture for abelian

varieties, using [EGFS25b].

Let $n \geq 5$ be an odd integer. Then the matroid $M^*(-W_n)$ is of rank $|E(W_n)| - |V(W_n)| = n - 1$ and contains $M(K_5)$ as a minor by Example 6.18, meaning it is not a cographic matroid because K_5 is not planar. This means that any admissible double cover of stable curves with irreducible components genus 0 and $\Sigma_\iota = -W_n$ gives a matroidal degeneration of Prym varieties $P^* \rightarrow (\Delta^*)^{2n}$ to a matroid which is not cographic, because $-W_n$ is tangled. The existence of such curves is established in Section 7.1. Then this family extends to an algebraic family $P^* \rightarrow B^* \supseteq (\Delta^*)^{2n}$ by [EGFS25a, Prop 4.10], giving us thus a map $B^* \rightarrow \mathcal{A}_{n-1}$ into the moduli space of principally polarized abelian varieties. Observe first of all that this map is not dominant when $n \geq 7$. Moreover, we can apply [EGFS25b, Thm. 1.6] to conclude that any very general fiber of the family fails the integral Hodge conjecture.

If $n \geq 6$ is even, start with $-W_{n-1}$ and add an edge between the center vertex and any vertex on the rim. By exactly the same argument as above, we obtain a degeneration of Prym varieties $P^* \rightarrow B^* \supseteq (\Delta^*)^{2n-1}$ such that any very general fiber fails the integral Hodge conjecture.

We will now finally prove Corollary 1.4 in the introduction.

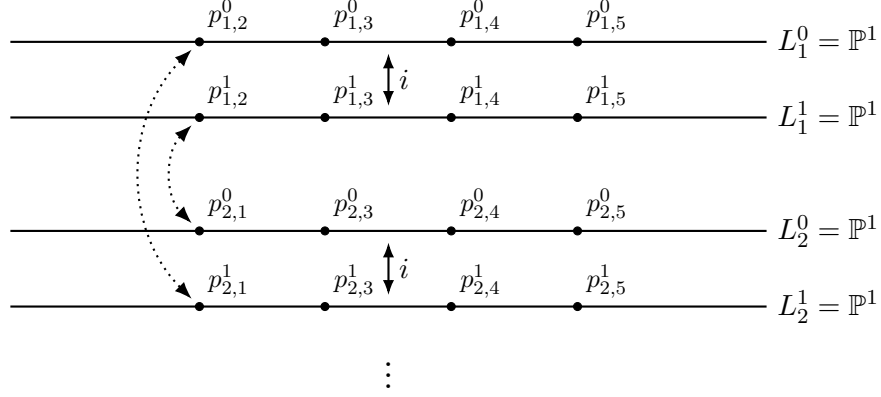
Corollary 7.13. *For $g \geq 6$, a very general point on the Prym locus $\mathcal{P}_g \subseteq \mathcal{A}_g$ fails the integral Hodge conjecture for one-cycles.*

Proof. Let $g \geq 6$. By [EGFS25b, Theorem 1.6], it suffices to find a matroidal degeneration of g dimensional Prym varieties $P^* \rightarrow B^*$ with underlying regular matroid not cographic. Let $\Sigma := (K_{6,g-5}^*, \sigma)$ be the signed graph associated to $K_{6,g-5}$, which is tangled (see Example 6.21 for details on this signed graph). Then by Theorem 7.9, there is a degeneration of Prym varieties $P^* \rightarrow (\Delta^*)^{3g}$ such that its monodromy cone is matroidal with regular matroid $M(K_{6,g-5})$. Moreover, the tangled signed graph Σ has at least three edges incident to each vertex because all faces of the graphs $K_{6,n}$ are triangles. Therefore, by the construction in the proof of Theorem 7.9, the admissible double cover of curves associated to the derived graph of Σ has irreducible components of genus 0. The degeneration is thus maximal. This means that the degeneration $P^* \rightarrow (\Delta^*)^{3g}$ is matroidal with regular matroid $M(K_{6,g-5})$, which is of rank g and not cographic, as it contains $M(K_6)$ as a minor. Then this family extends to an algebraic family $P^* \rightarrow B^* \supseteq (\Delta^*)^{3g}$ by [EGFS25a, Prop 4.10]. Because the dimension of the Prym locus $\mathcal{P}_g$ is $3g$, the induced map $B^* \rightarrow \mathcal{P}_g$ is dominant. We conclude by [EGFS25b, Thm 1.6.]. $\square$

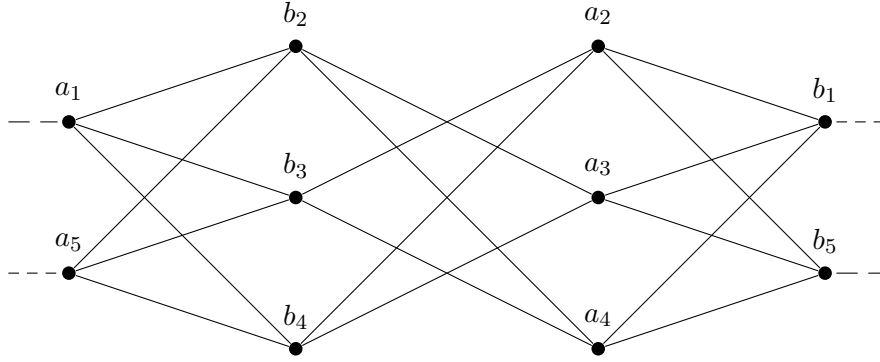
8. GWENA'S EXAMPLE

This chapter was an early exposition of Gwena's example before we discovered the theory of signed graphs. Almost everything here can be found in Gwena's thesis [Gwe04], except that the computations of the realization are slightly different: he uses the lattice $X^- = H_1(\Gamma(C_0), \mathbb{Z})/H_1(\Gamma(C_0), \mathbb{Z})^+$ whereas we use $\ker \delta$. They are related by the equality $\ker \delta = 2X^-$ (see [ABH01, pp. 27].

Suppose we have ten lines $\mathbb{P}^1$ labelled as L_i^ϵ with $1 \leq i \leq 5$ and $0 \leq \epsilon \leq 1$ and an involution $\tilde{\iota}_0$ on $\widetilde{C}_0 = \coprod_{i,\epsilon} L_i^\epsilon$ swapping each L_i^0 with L_i^1 . Mark distinct four distinct points on each line L_i^ϵ labeled as p_{ij}^ϵ with $j \neq i$. We identify the marked points $p_{ij}^0 \sim p_{ji}^1$:



Let the resulting space be C_0 and the involution descending to it ι_0 . The dual graph is depicted below. It should be noted that the vertices a_1 and b_5 are connected and b_5 is connected to a_1 , indicated by the dashes. The involution swaps a_i with b_i and acts on the edges according to the vertices.



The curve C_0 is clearly stable, ι_0 has no smooth fixed points and we only have non-fixed nodes. Thus this induces a degeneration of Prym varieties $P^* \rightarrow (\Delta^*)^{10}$. Label the edges for $i < j$ as follows:

$$e_{ij} := (a_i, b_j)$$

$$e'_{ij} := (b_i, a_j).$$

Take the spanning tree $T = (e_{12}, e_{13}, e_{14}, e'_{25}, e'_{34}, e'_{12}, e'_{13}, e'_{14}, e_{25})$. We list the basis for $H_1(\Gamma(C_0), \mathbb{Z})$ given by T in the table below.

$e \in H_1(\Gamma(C_0), \mathbb{Z}) \setminus T$	corresponding cycle in $H_1(\Gamma(C_0), \mathbb{Z})$
e_{15}	$\gamma_{15} = e_{15} + e_{13} + e'_{34} + e'_{14} - e'_{12} + e_{25}$
e'_{15}	$\gamma'_{15} = e'_{15} + e'_{25} - e_{12} + e_{13} + e'_{34} + e'_{14}$
e_{35}	$\gamma_{35} = e_{35} - e_{25} + e'_{12} - e'_{13}$
e'_{35}	$\gamma'_{35} = e'_{35} - e_{13} + e_{12} - e'_{25}$
e_{45}	$\gamma_{45} = e_{45} - e_{25} + e'_{12} - e'_{14}$
e'_{45}	$\gamma'_{45} = e'_{45} - e_{14} + e_{12} - e'_{25}$
e_{23}	$\gamma_{23} = e_{23} + e'_{12} - e'_{14} - e'_{34}$
e'_{23}	$\gamma'_{23} = e'_{23} + e'_{13} - e'_{14} - e'_{34} - e_{13} + e_{12}$
e_{24}	$\gamma_{24} = e_{24} + e'_{12} - e'_{14} - e'_{34} - e_{13} + e_{14}$
e'_{24}	$\gamma'_{24} = e'_{24} - e'_{34} - e_{13} + e_{12}$
e_{34}	$\gamma_{34} = e_{34} + e'_{13} - e'_{14} - e'_{34} - e_{13} + e_{14}$

and because the involution acts on the edges as $e_{ij} \mapsto -e'_{ij}$, we get the following identities:

$$\begin{aligned}
i(\gamma_{15}) &= -\gamma'_{15} - \gamma_{34} & i(\gamma'_{15}) &= -\gamma_{15} - \gamma_{34}, \\
i(\gamma_{35}) &= -\gamma'_{35} & i(\gamma'_{35}) &= -\gamma_{35}, \\
i(\gamma_{45}) &= -\gamma'_{45} & i(\gamma'_{45}) &= -\gamma_{45}, \\
i(\gamma_{23}) &= -\gamma'_{23} + \gamma_{34} & i(\gamma'_{23}) &= -\gamma_{23} + \gamma_{34}, \\
i(\gamma_{24}) &= -\gamma'_{24} + \gamma_{34} & i(\gamma'_{24}) &= -\gamma_{24} + \gamma_{34}, \\
i(\gamma_{34}) &= \gamma_{34}.
\end{aligned}$$

Thus a basis for $H_1(\Gamma(C_0), \mathbb{Z})^-$ is

$$\begin{aligned}
v_1 &= \gamma_{35} + \gamma'_{35} \\
v_2 &= \gamma_{45} + \gamma'_{45} \\
v_3 &= \gamma_{15} + \gamma'_{15} + \gamma_{23} + \gamma'_{23} \\
v_4 &= \gamma_{15} + \gamma'_{15} + \gamma_{24} + \gamma'_{24} \\
v_5 &= \gamma_{15} + \gamma'_{15} + \gamma_{34}
\end{aligned}$$

Moreover, because we are free on the vertices and edges, $\ker \delta = H_1(\Gamma(C_0), \mathbb{Z})^-$. Then the matroid realization $\varphi : E(\Gamma(C_0))/\iota \rightarrow (H_1(\Gamma(C_0), \mathbb{Z})^-)^\vee$ takes the shape

$[e] \in E/i$	$\varphi([e]) : H_1(\Gamma(C_0), \mathbb{Z})^- \rightarrow \mathbb{Z}$
$[e_{12}]$	$\begin{pmatrix} 1 & 1 & -1+1 & -1+1 & -1 \\ -1 & 0 & 1+1-1 & 1+1-1-1 & 1+1-1 \\ 0 & -1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -1 & -1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{pmatrix}$

giving us the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

One checks that this is a matrix representation of R_{10} using the following SageMath code:

```

A = Matrix(QQ, [
    [1,0,0,0,0, -1,1,0, 0,1],
    [0,1,0,0,0, 1, -1,1,0,0],
    [0,0,1,0,0, 0, 1, -1,1,0],
    [0,0,0,1,0, 0,0, 1,-1,1],
    [0,0,0,0,1, 1, 0,0, 1,-1]
])
B = Matrix(QQ, [
    [1, 0, 0, 0, 0, 1, -1, 0, 0, -1],
    [0, 1, 0, 0, 0, 1, 0, -1, 0, -1],
    [0, 0, 1, 0, 0, 0, 1, 0, 1, 1],
    [0, 0, 0, 1, 0, 0, 0, 1, 1, 1],
    [0, 0, 0, 0, 1, -1, 1, 1, 1, 1]
])
R10 = Matroid(matrix=A)
M = Matroid(matrix=B)
M.is_isomorphic(R10)

```

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